

Why Ellipse or Johannes Kepler, a Bigger Man

The No One Tells Us How To Think series

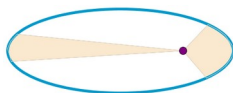
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Why Ellipse or Johannes Kepler, a Bigger Man

A Short Elementary Motivation For Further Study
Of Classical Celestial Mechanics

from the *No One Tells Us How Think* series

Roman Y. Andronov



To my wife, Tonya

Contents

Front matter: goals, prerequisites, target audience	1
Chapter 1, A Bigger Man	3
Chapter 2, The ABC Of Conic Sections	9
Chapter 3, Why Inverse Square	28
Chapter 4, Why Ellipse	36
Chapter 5, An Isomorphism In Celestial Mechanics	43

Goals, Prerequisites, Target Audience

This booklet has at least two goals. With a tiny body of work that you are holding in your hands we want to, first, introduce our readers to a more or less historically accurate, and compact, account of how Johannes Kepler (1571-1630) arrived at his results, known to posterity as the Kepler's Laws Of Planetary Motion, three in total.

That, first, part of the booklet represents its soft content.

Second, we want to show in as elementary of a fashion as possible into what the revolutionary ideas of Kepler blossomed several short hundreds of years later.

That, second, part of the booklet represents its hard-sciences content.

Technically, in that hard-sciences portion of the text we will prove the following two mutually converse implications, in that order:

- IF ellipse THEN inverse square and
- IF inverse square THEN ellipse

Our proof of the first implication above will be in the form of a mock discovery of the Universal Law Of Gravity. Our proof of the second implication above will use the basic machinery of differential and integral calculus, see below.

The goal of the final chapter of this small book is to wet the appetite of our readers further and, possibly, give them a chance to grow mathematically into classical celestial mechanics and come back and revisit that chapter later on, when their level of mathematical maturity catches up with the content presented in that chapter.

Prerequisites

The prerequisites for a successful journey through this text are reasonably low. The reader is only expected to be familiar with planar Euclidean geometry and basic differential and integral calculus at a level of a motivated upper high school or a proficient first-year-in-college student.

In addition, before we deduce the shape of the gravitational force based on the laws of planetary motion due to J. Kepler, we explicitly spell out an exhaustive list of initial assumptions, based on which our readers can judge what they need to know in order to comprehend these deductions.

Target Audience

From the goals of this booklet and its prerequisites it follows that there is virtually no age restriction to reading this text and, on the practical side, anyone between about 17 and 117 years of age who is interested in learning a basic mathematical explanation of why the planetary orbits are what they are is more than welcome to study this material.

But of course and as usual, there is no restriction on how much mathematics and physics a reader may already know. The more the merrier. However, the emphasis of this booklet is on an elementary exposition of the material in order to make the upcoming explanations accessible to as wide of an audience as possible.

By doing so we hope to motivate at least some of our readers to study the beautiful subject of classical celestial mechanics (much) further and (much) deeper.

Chapter 1

A Bigger Man

“Let me not seem to have lived in vain ...”

— Tycho Brahe’s last words

Tyge, Latinized in 1561 as Tycho, Brahe was born to Otte Brahe and Beate Bille on 14 December 1546 as a tip of a tip of a Danish nobility iceberg known as *Rigsraad* or Council of the Realm. An integral part of a closely knit oligarchy for several generations, the Brahes participated in seating and removing Danish kings, declaring wars, signing peace treaties and partitioning the kingdom into Geo-economic zones of influence that supplied the clan with an abundance of material wealth. In their time the Brahes *were* the fibers and the texture of the fabric called Denmark.

Though the people of Tycho’s social station were of *omnia mea mecum porto* kind, for their family name alone provided them with anything they would ever need in life, Tycho, winning the race out of his mother’s womb and leaving his twin brother behind in more than one way, bucked that trend at every major stage of his life.

Raised by his paternal uncle Jorgen Brahe and his wife Inger Oxe, Tycho accumulated a total of nine years of education across the universities of Copenhagen in Denmark and Wittenburg, Leipzig and Rostock in Germany despite the fact that masculine Jorgen was cut from a solid Viking warrior mold and was better with a sword than with a pen.

Not succumbing to family pressure, Tycho became a professional astronomer. Lost a good chunk of his nose in Rostock in a rapier duel with another Dane, named Manderup Parsberg, on 10 December 1566. Married, never officially, Kirsten Jorgensdatter, a woman of lower social standing. Designed his own instruments, including a meter-and-a-half long graduated compass and a quadrant that took forty grown men to install.

Over the arc of his life Tycho also spent 35 years and not a small fortune on tracking the motion of celestial bodies mostly from his observatory of Uraniborg on the island of Hven, within a short trip North of Copenhagen by a boat donated to *“our beloved Tyge Brahe”* in 1576 by Frederick II *“... to enjoy, use and hold; free and clear, without any rent ...”*

Tycho's revolutionary scientific contribution, and a silent message to all the talking heads of his time and those who came before him, was:

stop wagging your tongue, start measuring

meaning, do produce hard experimental data and produce he did.

At 16 years of age, in August of 1563, driven by the ambitious goal of achieving a minute of arc, $1/60$ -th of a degree or $\pi/10800$ -th of a radian, or better precision of measurements, Tycho started his first notebook on astronomical observations, a humble beginning that many years and publications later took him to Prague where in early 1600 his orbit and that of Johannes Kepler finally intersected due to the forces of religious fervor, political power struggle and intrigue that acted on Tycho and the forces of sheer madness of the Thirty Years' War that acted on Kepler.

In comparison with Tycho's societal pedigree, Kepler was a rather unsophisticated simpleton who was seen as a rival and treated with suspicion by and received a piece-meal treatment from his older and more famous patron. One measly planetary coordinate during the meal time today, another one - tomorrow. What a pity! A slow-walking of hard data that only delayed the inevitable.

The greatest experimentalist and the greatest theoretician of the time, Tycho and Kepler, were locked in a back and forth bickering duel fed by mutual mistrust, the battle that led each one, separately, into nowhere. For reasons unknown to us today the idea of *sharing* the spoils of a major discovery was not acceptable for these great men.

After Tycho's death in October of 1601, Kepler, now an official Imperial Mathematician, managed to gain full access to the observational data set from the stubbornly resistant Brahes, who sought to monetize the data, and promptly launched a three-year fitting numbers into a curve calculation spree during which Kepler had to face one crushing blow after another.

For one, Tycho's numbers killed Kepler's Platonic Solids idea dead and, eventually, the prevailing circular orbits theory.

Call it a happy coincidence, but out of all the known planets Tycho charged Kepler with investigating the only one whose orbit was perceptibly different from a circle, the troublesome, looping Mars. Not one

but two of Tycho's observations were off by eight minutes of arc from Kepler's circular orbit candidate which meant that there was no room for circles and spirals in planetary astronomy. Thus, Kepler was left, as he succinctly put it himself, with *"only a single cartful of dung"*.

Possibly out of despair, Kepler ran the numbers against an ellipse (*an oval of some kind*) and ... drum roll ... it worked perfectly!

Here we must give Kepler all the credit that is due for having the guts to face and admit the error of his ways, for having the courage to go against and brush off the popular opinion and the dogma of preconceived notions when in 1609 in a quick one-two punch fashion he put together the first two laws of planetary motion:

- a planet traces an ellipse with the Sun in one of its two foci (**K1**) and
- with respect to the Sun the planet's radius vector sweeps the sectors of equal areas in equal times (**K2**)

In modern physics Kepler's second law is called the law of constancy of areal velocity which follows from the law of conservation of angular momentum.

Regardless, however, we have no choice but to admit that despite of all of his follies, Johannes Kepler was a bigger man, especially given the overall, grand, academic atmosphere of his time. It took quite a bit of courage to bravely step into the unknown. Alone.

Ten years later, in 1619, closing the **"compute it"** chapter of his life, Kepler formulated the third and the final law of planetary motion:

- the square of a planet's period of revolution about the Sun is proportional to the cube of its average distance from the Sun (**K3**)

Kepler passed away in 1630 and the world had to wait for another 36 (according to Newton's 1686 letter to Edmund Halley) years for the vindication of all the work carried out by him and Tycho.

In 1687 Isaac Newton, turning Galileo's **"explain it mathematically"** idea into a reality, published *"Principia"* where in mostly geometric form, not in the modern calculus notation, in an ensemble of propositions he deduced the law that later became known as the Universal Law of Gravity with a notion of a central force at its core.

In modern mathematical terms of conditional/implication construct, applied to planetary motion, we have:

if the Universal Law of Gravity holds then a planet traces an ellipse with the Sun at one of its two foci

Textbooks upon textbooks on undergraduate physics courses throughout the world are filled with the proof of the above statement and we shall touch upon it later on.

However, what we are really interested in is a plausible deduction of the above statement's more historically accurate converse:

if a planet traces an ellipse with the Sun at one of its two foci then the Universal Law of Gravity holds

In the earlier mentioned 1686 letter to Edmund Halley Newton admits *"I can affirm that I gathered it from Kepler's theorem about twenty years ago"*.

For historical accuracy sake, it must be said that other physicists, independently of Newton, also came up with the idea of a central force inversely proportional to certain powers of the distance: Robert Hooke (1635-1703) , Christiaan Huygens (1629-1695), Edmund Halley (1656-1742) and Christopher Wren (1632-1723).

Even Kepler, who foretold the manned space flights, writing about the celestial machine posited that *"... nearly all the manifold movements are carried out by means of a single, quite simple magnetic force"*.

In a letter to Newton Robert Hooke mentioned the inverse *square* idea, while in 1659 Christiaan Huygens deduced the inverse *first* power of distance law applicable to a point mass tracing a perfect circle:

$$F_c = \frac{mv^2}{r}$$

However, Hooke admitted that he did not know how to compute the trajectories traced by the bodies acted on by a central force and Huygens, adhering to the idea that a body begins to move only because another body bumped into it, was unwilling to subscribe to the action-at-a-distance concept.

Huygens obviously did entertain an idea of a central force acting on an apple, but only on an apple spinning in a circle *on a rope* because the rope was tangible, clear and present and easily understood.



If there is no rope tying Mars to the Sun then the only reason why Mars does what it does is because (exaggerating for demonstration purposes) myriads of tiny elves flapping their tiny wings push the planet around the Sun constantly adjusting their course in a perfectly and precisely choreographed effort.

An infinite number of apples could have showered Huygens for eternity but, being a rational philosopher, he would have never done what Newton did.

After (our guess) unbuckling his thought process from the shackles of the popular opinion, in a wild mix of skepticism, intuition and insight, Newton proposed exactly the opposite:

- there **is** this mysterious thingy dubbed *central force due to gravity* that:
 - acts at a distance
 - does not need a medium
 - is not visible to the human eye
- the same, gravitational, force rules the motion of *all* the planets
- (as apparently became obvious to Newton when an apple fell to his feet in Woolsthorpe) the same force also rules the motion of the planets's moons, the same force acts on apples falling from the apple trees as well as on all material objects, in other words, that force is *universal*

The quick challenge to explain what causes gravity and how exactly its action is propagated was settled by Newton with "*for us it is enough*", meaning that a precise enough mathematical description of the manifestation of the force is all that is needed in order to explain the observable behavior of macro objects, the planets and their moon, for example.

In order to put Newton's courage and genius into a historical perspective, take the currently standing wisdom of "*time travel is impossible*", for example, and come out in public and state that it **is** possible and then back it up with a credible mathematical apparatus verifiable by an experiment.

Even the small number of paragraphs at the beginning of this chapter demonstrates that scientific developments as they unfold in real time, carried out by people who are not armed with the ahead of time knowledge, span multiple generations, time, and countries, space, and make up an incredibly complex process that is not and, in all fairness, cannot and, ruthlessly enough, should not be at the core of raw academic undergraduate curricula.

Pedagogically the above process is compressed into a conveniently tiny package of a side bar or a small print at the bottom of a page, at best. Most of the time, however, even that courtesy is not extended to the pioneers of the results being covered.

The fact that that compression unavoidably leads to an accepted loss of historical accuracy tells us that we care more about the *what* and not the *who* unless, of course, it comes to *us* getting the credit that we, of course, deserve.

As such, not only because we cannot possibly get into another person's mind, but mostly in order to avoid the idle speculation, we present here a plausible way of how *we* and *nowadays* can arrive at the above theorem admitting, of course, that we are armed with the after the fact knowledge, specific education in the field and so on.

Before we lay out our first proof, we spell out exactly and exhaustively all and the only ingredients that we are allowed to use for the purposes of a mock discovery.

As we conclude the soft content part of our booklet and before we move on to its hard content, we would like to spell out the key takeaways of this chapter.

1. Tycho Brahe (1546-1601) may or may not have kick-started hard sciences in Western Europe, but he definitely advanced them greatly because his actions, producing mountains of hard experimental data, created a foundation of these sciences: *stop speculating, start measuring*
2. Johannes Kepler (1571-1630) was, perhaps, one of the first theoretical physicists because he took Galileo's mandate of "*explain these measurements mathematically*" to fruition. Kepler also was a bigger man because he had the guts and the courage to admit the error of his ways and bravely step into the unknown, all by himself

Chapter 2

The ABC Of Conic Sections

In the upcoming solution of a certain differential equation in Chapter 4 we will arrive at an expression that we will claim to be precisely an equation of a conic section. Because in this case a single equation covers an entire family of several different planar curves, in this chapter we explain exactly how and why does this phenomenon come into existence.

In particular, it so happens that a single mathematical equation mentioned above describes the following curves also known as *conic sections*:

- a parabola
- a hyperbola and
- an ellipse

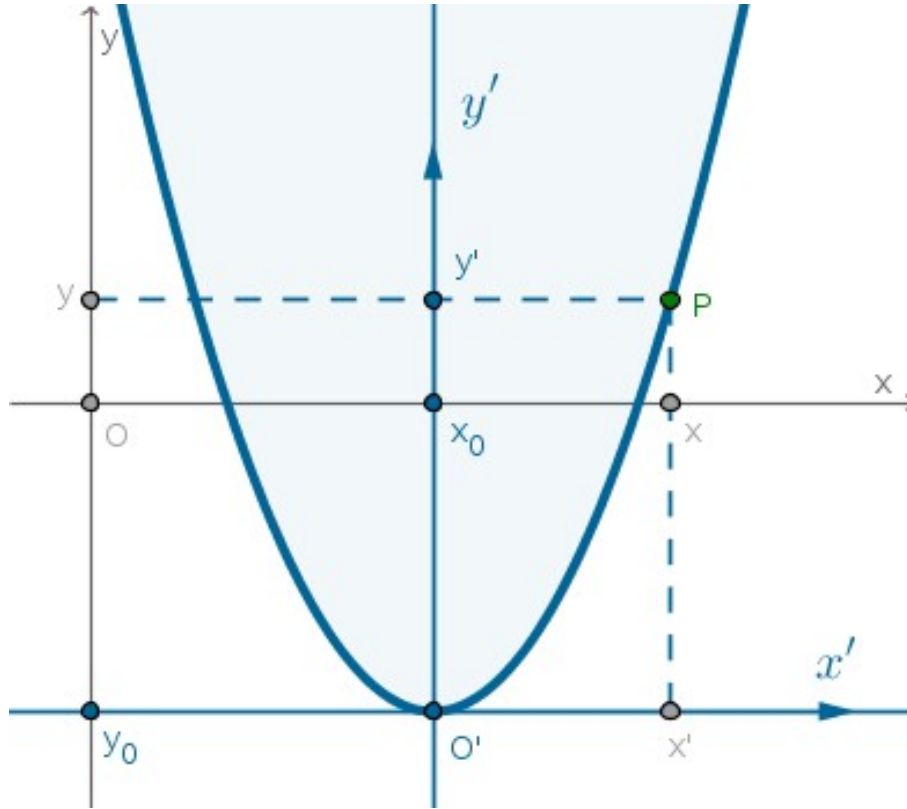
While, admittedly, another popular curve, *a circle*, does not really fit into the framework of the said equation neatly, it is still possible to inscribe a circle into the happy family of conic sections if we agree to define the focal parameter of a circle as its radius and if we set the eccentricity of a circle to zero, see more on that below.

2.1 Parabola

Recall that a graph of a quadratic polynomial ($a, b, c \in \mathbb{R}$ with $a \neq 0$):

$$y(x) = ax^2 + bx + c$$

is a *parabola* or a curve that has a single vertex, O' , that can be defined intuitively as a point of largest curvature or as a point where parabola's two horns meet (Figure 2.1):



Assuming that in a given Cartesian **C**oordinate **S**ystem, or CS for short from now on, xOy a parabola's graph is not positioned conveniently, can we find a new equivalent CS $x'O'y'$ in which the parabola keeps its shape and orientation but still has the point O' as its vertex? If such a CS exists then for any point P on the curve it must be true that:

$$x = x_0 + x' \quad \text{and} \quad y = y_0 + y'$$

If we put these expressions for x and y back into the original expression of the quadratic polynomial then we will have:

$$y_0 + y' = a(x_0 + x')^2 + b(x_0 + x') + c$$

$$y' = ax_0^2 + 2ax_0x' + ax'^2 + bx_0 + bx' + c - y_0$$

Collect all the like x' terms above:

$$y' = ax'^2 + (2ax_0 + b)x' + (ax_0^2 + bx_0 + c - y_0)$$

and *require* that the coefficient in front of x' and the free floating term above) must be zero:

$$2ax_0 + b = 0 \quad \text{and} \quad ax_0^2 + bx_0 + c - y_0 = 0$$

Solve the leftmost equation above for x_0 :

$$x_0 = -\frac{b}{2a}$$

and use that expression for x_0 in order to solve the rightmost equation above for y_0 :

$$y_0 = -\frac{b^2}{4a} + c$$

We, thus, have found a new CS where:

$$y' = ax'^2$$

and which is obtained from the original one by shifting the new origin, O , by x_0 along the x -axis and by y_0 along the y -axis.

Omitting the primes in the above equation and switching the names of the axes around, by naming the old y the new x and by naming the old x the new y or simply rotating our parabola in CS's plane about its vertex clockwise by ninety degrees, we arrive at the following equation:

$$y^2 = \frac{1}{a} x$$

What is the meaning of this awkward factor $1/a$ in front of x above? Recall the notion of a *fixed point* from (technically) mathematical analysis: it is a point for which numerically the argument and the return value of the function match. For example, for the relation $y = x^2$ the fixed point is $(1, 1)$ because when $x = 1$ then $y = |1|$.

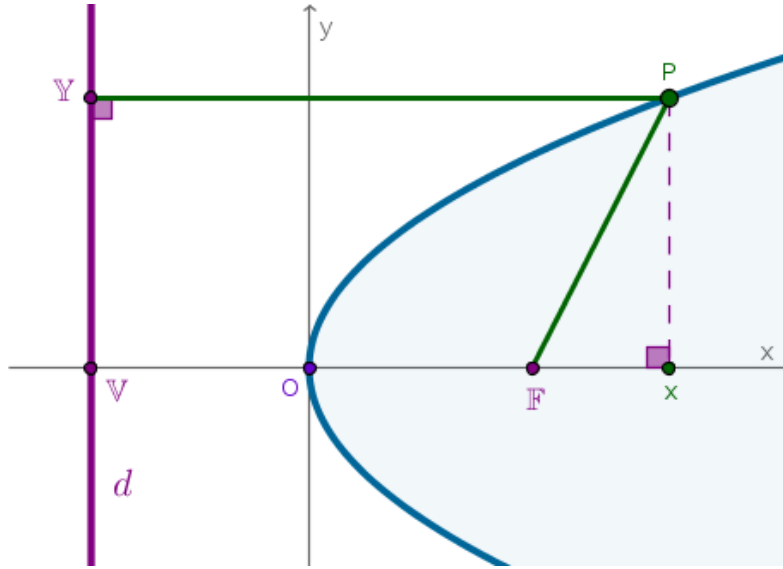
Geometrically it means that strictly vertical and horizontal distances from such a point on the graph of a function to the corresponding axes are equal.

Does this condition hold for any and all the points on our parabola?

Obviously not.

Well, what if we relax the strict vertical/horizontal requirement a bit and instead ask for the distance from any point, P , on the curve to a fixed point, $\mathbb{F}$, on the x axis to be equal to the distance from the

said point P to a point, $\mathbb{Y}$, that is located on the shifted image, d , of the axis y located a certain amount, $\mathbb{V}$, of units away from the point O (Figure 2.2):



In other words, we require that:

$$|PY| = |PF|$$

$$|OV| + x = \sqrt{(x - |OF|)^2 + y^2}$$

But we already agreed that $y^2 = (1/a)x$. Hence:

$$|OV| + x = \sqrt{x^2 - 2x|OF| + |OF|^2 + \frac{1}{a}x}$$

Is there a simple condition under which the above equality holds? Well, as we answer that question, let us ponder about what will happen if the expression under the above square root collapses into a perfect square $(x + |OF|)^2$? For that to occur we should have:

$$-2x|OF| + \frac{1}{a}x = 2x|OF|$$

which implies that:

$$\frac{1}{a} = 4 \cdot |OF|$$

in which case for our main expression with the square root above we must have:

$$|OV| + x = x + |OF|$$

or:

$$|OV| = |OF|$$

which means that for our wishful thinking to be true the parabola's vertex, O , must be located half way between the point F and the straight line d . It follows then that if the ratio $1/a$ is interpreted as four distances from the magic point F to the parabola's vertex, O , then the relation:

$$|OV| + x = \sqrt{x^2 - 2x|OF| + |OF|^2 + \frac{1}{a}x}$$

holds and our curve, a parabola, may be re-defined as:

a parabola is a locus of points each of which is equidistant from a fixed straight line, d , called the **directrix**, and a fixed point, F (not on d), called the parabola's **focal point**

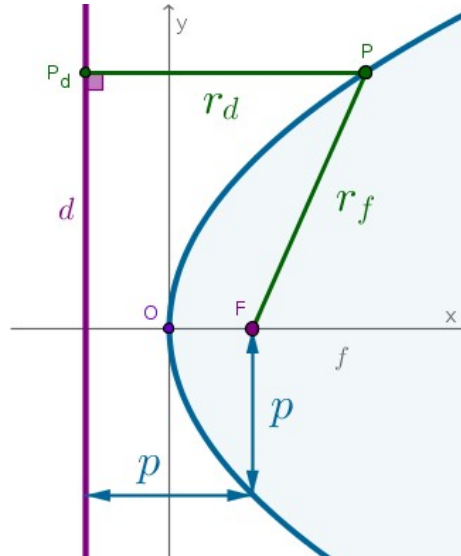
In order to get rid of the awkward factor $1/a$ above, let us name the distance $|FV|$ as p . Then from our earlier demand that $1/a = 4 \cdot |OF|$ it follows that:

$$\frac{1}{a} = 4 \cdot |OF| = 4 \cdot \frac{p}{2} = 2p$$

and we arrive at the parabola's *canonical equation* in its *canonical CS*:

$$y^2 = 2px$$

where the real number p is called the curve's *focal parameter* and the unique straight line passing through the curve's vertex and its focal point is called the *focal axis*, f (Figure 2.3):



If we put the focal point's x -coordinate of $p/2$ back into the parabola's canonical equation:

$$y^2 = 2p \frac{p}{2} = p^2 \implies y = \pm p$$

then we obtain yet another and a more reusable definition (or interpretation) of the curve's focal parameter p : it is one half of the length of the *focal chord* or a line segment trapped between the curve's two horns and constructed as a perpendicular to that curve's focal axis, f , through the curve's focal point, $\mathbb{F}$.

A parabola has a plethora of interesting properties but, in setting the stage for the remaining two curves, we are interested in the following one. If we name the distance between an arbitrary point, P , on the curve and its focal point, $\mathbb{F}$, as r_f :

$$r_f = |P\mathbb{F}|$$

and if we name the distance between such a point P and the curve's directrix, d , as r_d :

$$r_d = |PP_d|$$

then without much effort, since the ratio between the distances r_f and r_d is the unity by definition:

$$\boxed{\frac{r_f}{r_d} = \text{const}(= 1)}$$

we arrive at an interesting pattern, which, as we shall discover soon enough, holds for hyperbolas and ellipses as well:

a parabola is a locus of points for each of which the above ratio is a specific non-negative real constant number equal to 1 (in this case)

Naming the distance between the curve's focal point and its directrix as r_{fd} and using the letter **p** for a parabola as a superscript, we have the following symbolic definition of a parabola:

$$\boxed{r_{fd}^p = p}$$

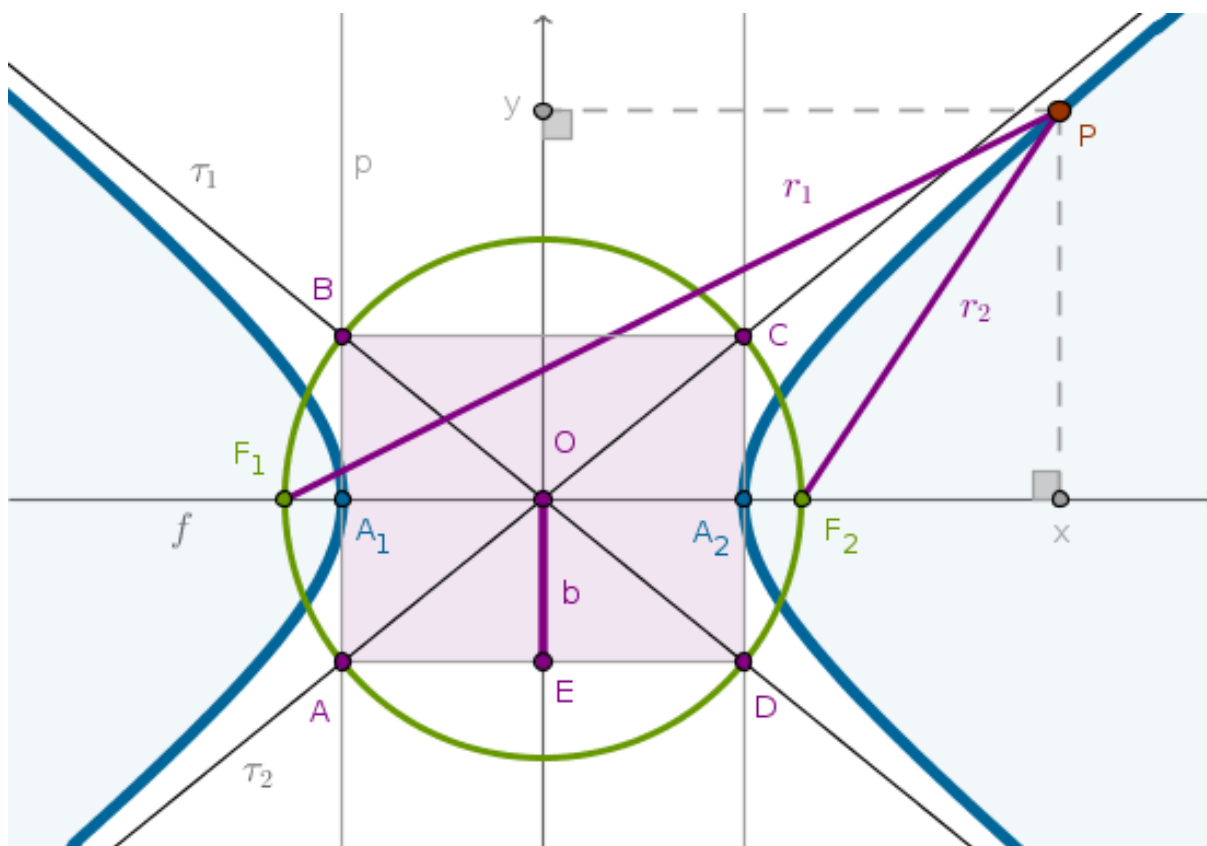
which, in its structure, will hold up for the like definitions of a hyperbola and of an ellipse.

2.2 Hyperbola

We observe in passing that a number of popular, planar or otherwise, curves that middle and/or high school students may be introduced to can have several different, but equivalent, types of definitions: algebraic, geometric, mechanical or optical.

For example, a popular curve known as a *cycloid* may have a purely mechanical origin because such a curve can be defined as follows. A cycloid is a curve that is traced by a single point located on the periphery of a circle that rolls along a straight line with sliding or spinning.

Thus, according to one of its geometric definitions, our next conic section, a *hyperbola*, is a locus of coplanar points the absolute value of the difference of the distances from each of which to two fixed distinct points located in the same plane, called the hyperbola's **foci** (the points F_1 and F_2 in the diagram blow), is a constant positive real number, traditionally designated as $2a = OA_1 = OA_2$ (Figure 2.4):



The half of the distance $F_1F_2 = 2c$, ($c > 0, c \in \mathbb{R}$) is called the hyperbola's **focal distance**, the unique straight line f passing through the points F_1 and F_2 is called the hyperbola's **focal axis** (f) and one half of the length of the line segment A_1A_2 , or a , where $A_{1,2}$ are the points at which both branches of a hyperbola intersect the x -axis, is called the hyperbola's **major semi-axis**.

By definition an arbitrary point $P(x, y)$ belongs to the graph of a hyperbola iff:

$$|r_1 - r_2| = 2a \quad \text{or} \quad r_1 - r_2 = \pm 2a$$

From the right triangles F_1xP and F_2xP for r_1 and r_2 we have:

$$\sqrt{(x + c)^2 + y^2} - \sqrt{(x - c)^2 + y^2} = \pm 2a$$

Carry the second radical from the above left hand side to its right hand side:

$$\sqrt{(x+c)^2 + y^2} = \pm 2a + \sqrt{(x-c)^2 + y^2}$$

and square both sides of the result:

$$(x+c)^2 + y^2 = 4a^2 \pm 4a\sqrt{(x-c)^2 + y^2} + (x-c)^2 + y^2$$

After canceling the like terms above, we have:

$$4cx = 4a^2 \pm 4a\sqrt{(x-c)^2 + y^2}$$

$$cx - a^2 = \pm a\sqrt{(x-c)^2 + y^2}$$

Square both sides of the last equality:

$$c^2x^2 - 2cxa^2 + a^4 = a^2x^2 - 2a^2cx + a^2c^2 + a^2y^2$$

Cancel the $2cxa^2$ terms above and group the terms with x^2 and a^2 together in order to obtain:

$$c^2x^2 - a^2x^2 - a^2y^2 = a^2c^2 - a^4$$

$$(c^2 - a^2)x^2 - a^2y^2 = a^2(c^2 - a^2)$$

Since the numbers c and a are constant, so is the number $c^2 - a^2$. From the triangle F_1PF_2 and the triangle inequality it follows that:

$$F_1F_2 \geq PF_1 - PF_2$$

which means that:

$$c \geq a$$

We, of course, can consider the strict equality above but it is not interesting as it corresponds to the locus of points that satisfy:

$$F_1F_2 = PF_1 - PF_2 \quad \text{and} \quad F_1F_2 = PF_2 - PF_1$$

which, barring the line segment F_1F_2 , make up the entire straight line f and none of these points obviously belong to our curve. As such, it is safe to only consider the cases when:

$$c > a$$

which means that the difference $(c^2 - a^2)$ is strictly positive which, further, means that there always exists a real number b such that: $b^2 = c^2 - a^2$.

Putting that shape of the number b into our last equality above, we obtain:

$$b^2 x^2 - a^2 y^2 = a^2 b^2$$

or, after dividing both sides of the result by $a^2 b^2$, which is allowed since neither a nor b are zero, we have:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

or the hyperbola's *canonical equation* in its *canonical CS*. In order to better understand what that number b actually means geometrically, solve the last equation for y :

$$y = \pm \frac{b}{a} \sqrt{x^2 - a^2}$$

When x (by magnitude) becomes very very large then the magnitude of a^2 above can be neglected and we obtain the equation of the hyperbola's asymptote (τ_1 and τ_2 in the drawing above):

$$y = \pm \frac{b}{a} x$$

from where it follows that when $x = a$ then:

$$y = \pm b$$

and we arrive at a geometric interpretation of the number b (the hyperbola's **minor semi-axis**):

construct the hyperbola's both asymptotes τ_1 and τ_2 , construct a perpendicular p to f through the point A_1 (which is a straight line $x = -a$) in order to locate the points A and B , the points of intersection of the perpendicular p with the asymptotes.

Locate the points C and D likewise. Construct the hyperbola's **main rectangle** $ABCD$ and one half of its height will be exactly $b = OE$. We observe in passing that all six points A, F_1, B, C, F_2 and D are cocircular, since they belong to the circumference of the circle with the center at the point O and the radius $c = OF_1 = OF_2$.

We now define the curve's **eccentricity** as a constant real number e such that:

$$e = \frac{c}{a} > 1$$

(because $c > a$) and use it in order to express the distances r_1 and r_2 in our very first equation with radicals above in terms of a, e and x . If an arbitrary point $P(x, y)$ is on the graph of a hyperbola then for the distance r_1 we have:

$$r_1 = \sqrt{(x + c)^2 + y^2}$$

and then:

$$y^2 = \frac{c^2}{a^2}x^2 - x^2 - c^2 + a^2 = e^2x^2 - x^2 - c^2 + a^2$$

$$r_1 = \sqrt{x^2 + 2xc + c^2 + e^2x^2 - x^2 - c^2 + a^2} = \sqrt{e^2x^2 + 2xc + a^2}$$

But earlier we agreed that $c = a \cdot e$:

$$r_1 = \sqrt{e^2x^2 + 2aex + a^2} = \sqrt{(a + ex)^2}$$

which means that:

$$r_1 = \pm(a + ex)$$

and in a similar fashion we find that:

$$r_2 = \pm(a - ex)$$

Since the distances r_1 and r_2 must be positive real numbers, the signs in these expressions must be carefully chosen. From the canonical equation of hyperbola it follows that $|x| \geq a > 0$ and, since $e > 1$, in general we have $|ex| > a$ which means that when $x > 0$ then the magnitude in parenthesis in $r_1 = \pm(a + ex)$ is positive and the sign in front of these parenthesis must a plus:

$$r_1 = a + ex$$

while the magnitude in parenthesis in $r_2 = \pm(a - ex)$ is negative and the sign in front of these parenthesis must be a minus:

$$r_2 = -a + ex$$

Via the same reasoning we can show that when $x < 0$ then:

$$r_1 = -a - ex$$

and:

$$r_2 = a - ex$$

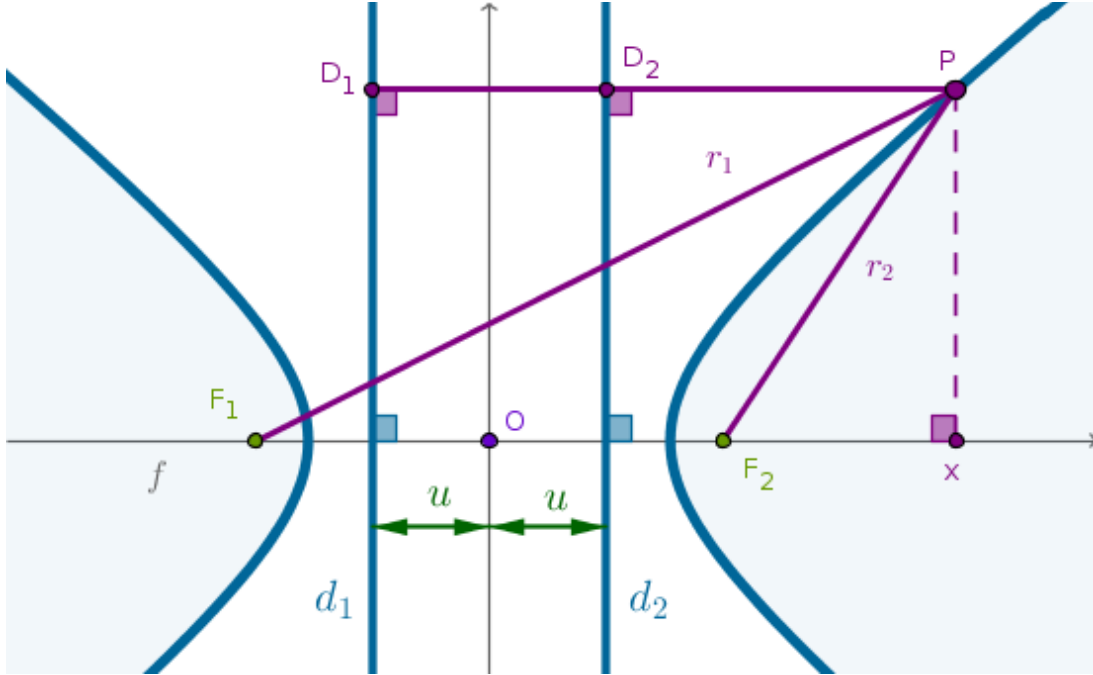
and in any case we arrive back at our original expression that started this investigation in the first place:

$$r_1 - r_2 = \pm 2a$$

The next question we have is this. If a *parabola* has a directrix then what about a *hyperbola*?

Assuming that a hyperbola, which has two horns, has two directrices, d_1 and d_2 , where would they be located?

Well, the hyperbola's directrices, shown in solid blue on the diagram below, should be positioned (Figure 2.5):



an unknown to us yet u units away from the origin, O , in such a way, as we alleged earlier in the parabolas relation $r_{fd}^p = p$, that:

$$\frac{r_1}{PD_1} = \text{const} = \frac{r_2}{PD_2}$$

Taking the distances r_1 and r_2 from their algebraic definitions, $r_1 = a + ex$ and $r_2 = -a + ex$, and using the fact that:

$$PD_1 = x + u, \quad PD_2 = x - u$$

where we take u to be a distance or a real positive number, we obtain:

$$\frac{a + ex}{x + u} = \frac{-a + ex}{x - u}$$

Solve the above equation for u :

$$(a + ex)(x - u) = (-a + ex)(x + u)$$

$$ax - au + ex^2 - exu = -ax - au + ex^2 + exu$$

$$2ax = 2exu \implies u = \frac{a}{e}$$

We, thus, find that the hyperbola's directrices are two straight lines, d_1 and d_2 perpendicular to the hyperbola's focal axis, f , and located $\frac{a}{e}$ units away from the canonical CS's origin, O , and in that CS the hyperbola's directrices have the following equation(s):

$$x = \pm \frac{a}{e} = \pm \frac{a^2}{c}$$

Use the value of $u = a/e$ in order to verify that our $r_f/r_d = \text{const}$ parabola's hypothesis holds for a hyperbola, *taking care to match the curve's focal point to the corresponding directrix*:

$$\frac{r_f}{r_d} = \frac{|a + ex|}{\left|x + \frac{a}{e}\right|} = e = \text{const}$$

In order to compute the distance between this curve's focal point and the corresponding directrix, we introduce the hyperbola's focal parameter, p , in the same, parabolic, fashion as one half of the length of the curve's focal chord which we instantly obtain from $b^2 = c^2 - a^2$ and the hyperbola's canonical equation by taking $x = c$:

$$y^2 = \frac{b^2 c^2}{a^2} - \frac{b^2 a^2}{a^2} = \frac{b^2}{a^2}(c^2 - a^2) = \frac{b^4}{a^2}$$

from where:

$$|y| = p = \frac{b^2}{a}$$

And now, for a hyperbola's version of r_{fd} from the above results we have:

$$r_{fd}^h = c - u = c - \frac{a^2}{c} = \frac{c^2 - a^2}{c} = \frac{b^2}{ae} = \frac{b^2}{a} \cdot \frac{1}{e}$$

$$\boxed{r_{fd}^h = \frac{p}{e}}$$

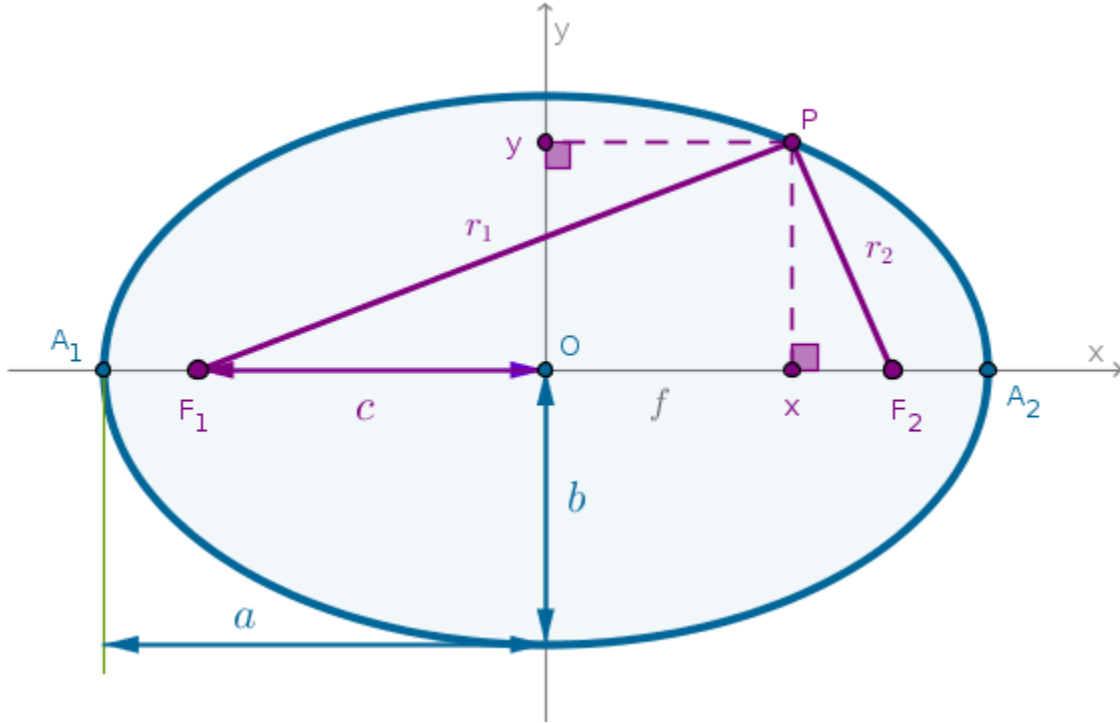
which is what we promised to be true earlier.

2.3 Ellipse

We shall model our elliptic steps directly after their hyperbolic counterparts which we just carried out and, as such, we will move a bit faster.

According to one of its geometric definitions, *an ellipse* (not degenerated into a circle) is a locus of (2-space) points the sum of the distances from each of which to two fixed distinct points, called **foci** (the

points F_1 and F_2 in the diagram below), is a constant positive real number, traditionally designated as $2a = OA_1 = OA_2$ (Figure 2.6):



The half of the distance $F_1F_2 = 2c$, ($c > 0, c \in \mathbb{R}$) is called the ellipse's **focal distance**. The unique straight line, f , passing through the points F_1 and F_2 is called the ellipse's **focal axis**. One half of the length of the line segment A_1A_2 , or a , is called the ellipse's **major semiaxis**.

By definition, an arbitrary point $P(x, y)$ belongs to the graph of an ellipse iff:

$$r_1 + r_2 = 2a$$

From the right triangles F_1xP and F_2xP for the distances r_1 and r_2 we have:

$$\sqrt{(x+c)^2 + y^2} + \sqrt{(x-c)^2 + y^2} = 2a$$

Carry the second radical on the left hand side of above to its right hand side and square the resulting equation *twice*, in order to obtain:

$$(a^2 - c^2)x^2 + a^2y^2 = a^2(a^2 - c^2)$$

From the triangle F_1PF_2 and the triangle inequality it follows that:

$$F_1F_2 = 2c \leq PF_1 + PF_2 = 2a$$

which implies that $a \geq c$.

However, the above case of the strict equality is not interesting because it corresponds to the locus of points P such that:

$$F_1F_2 = PF_1 + PF_2$$

which make up the line segment F_1F_2 . As such, we insist that for an ellipse:

$$a > c$$

Hence, there always exists a (positive) real number, b , such that:

$$b^2 = a^2 - c^2$$

Using that definition of the real number b in the last equation in x and y , we obtain the ellipse's canonical equation in its canonical CS:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

where geometrically the number b , we now understand, corresponds to the ellipse's **minor semiaxis**.

Defining the ellipse's **eccentricity** as a constant real number, e , such that:

$$e = \frac{c}{a} < 1$$

we use it to express the above distances r_1 and r_2 in $r_1 + r_2 = 2a$ in terms of a , e and x . Namely, if an arbitrary point $P(x, y)$ is on the graph of an ellipse then for r_1 we have:

$$r_1 = \sqrt{(x + c)^2 + y^2}$$

From the definition of the number b and the canonical equation of an ellipse for y^2 we have:

$$y^2 = a^2 - c^2 - x^2 + \frac{c^2}{a^2}x^2$$

Thus, for the distance r_1 we have:

$$r_1 = \sqrt{a^2 + 2xc + \frac{c^2}{a^2}x^2} = \sqrt{a^2 + 2aex + e^2x^2}$$

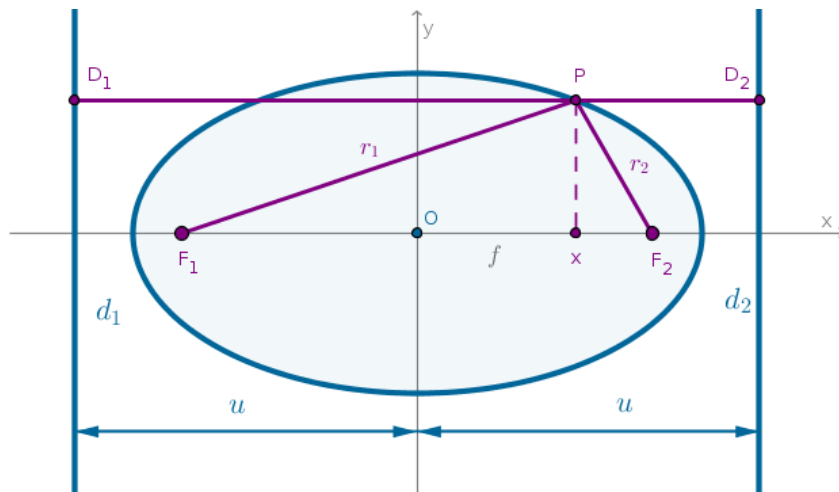
$$r_1 = \sqrt{(a + ex)^2} = \pm(a + ex)$$

But since $a > c$, it follows that $|x| \leq a$ and from the definitions of the numbers b and c we have $0 < e < 1$ which means that in general $|ex| < a$ and that $r_1 = a + ex$ and, by analogy, for the distance r_2 we obtain: $r_2 = a - ex$.

And we, thus, arrive back at our original, geometric, definition of an ellipse: $r_1 + r_2 = 2a$.

Where would the ellipse's two directrices, d_1 and d_2 , be located?

Let us think about this together (Figure 2.7):



According to our working hypothesis, they should be located u units away from the CS's origin, O , such that:

$$\frac{r_1}{PD_1} = \text{const} = \frac{r_2}{PD_2}$$

But:

$$PD_1 = x + u, \quad PD_2 = u - x$$

and hence:

$$\frac{a + ex}{x + u} = \frac{a - ex}{u - x}$$

Solve the above equation for u :

$$u = \frac{a}{e}$$

We, thus, find that the ellipse's directrices have the same analytic form as their hyperbolic counterparts (in respective canonical Css):

$$x = \pm \frac{a}{e} = \pm \frac{a^2}{c}$$

and, use the magnitude of $u = a/e$, we verify that our grand hypothesis holds for an ellipse *taking care to match the curve's focal point to the corresponding directrix*:

$$\frac{r_f}{r_d} = \frac{|a + ex|}{\left|x + \frac{a}{e}\right|} = e$$

In order to compute the distance between this curve's focal point and the corresponding directrix, we introduce the ellipse's **focal parameter**, p , in the same, parabolic, fashion as one half of the length of the curve's focal chord which we instantly obtain from the definition of b , as $b^2 = a^2 - c^2$, and from the ellipse's canonical equation by setting $x = c$:

$$y^2 = \frac{b^2}{a^2}(a^2 - c^2) = \frac{b^4}{a^2} \quad \text{or} \quad |y| = p = \frac{b^2}{a}$$

As such, for an ellipse's version of r_{fd} we have:

$$r_{fd}^e = u - c = \frac{a^2}{c} - c = \frac{a^2 - c^2}{c} = \frac{b^2}{ae} = \frac{b^2}{a} \cdot \frac{1}{e}$$

or:

$$\boxed{r_{fd}^e = \frac{p}{e}}$$

And now, that we hopefully have a better understanding of the parameters p and e for all three curves, we can go back to a parabola and define *its* eccentricity as the positive unity:

$$e^p = 1$$

in which case the respective equations for *a parabola* become:

$$\frac{r_f}{r_d} = e(= 1) \quad \text{and} \quad r_{fd}^p = \frac{p}{e}$$

If we omit the qualifying superscripts in all of these equations then for all three curves we have:

$$\frac{r_f}{r_d} = e \quad \text{and} \quad r_{fd} = \frac{p}{e}$$

So far we've proved:

IF a point, P , belongs to a curve THEN certain properties hold

statements.

Conversely, even though we omit them:

IF certain properties hold THEN a point, P , belongs to a curve

statements can be proved. It is, thus, can be shown that the class of the curves that depict parabolas, hyperbolas and ellipses can be defined as follows:

each curve of that class (and only the curve of that class) is a locus of points for which $r_f = r_d \cdot e$ where e is a positive constant real number; the curve is an ellipse if $e < 1$, a parabola if $e = 1$ and a hyperbola if $e > 1$

and we did not show the proof of that part in order to keep the size of this booklet comprehensibly small.

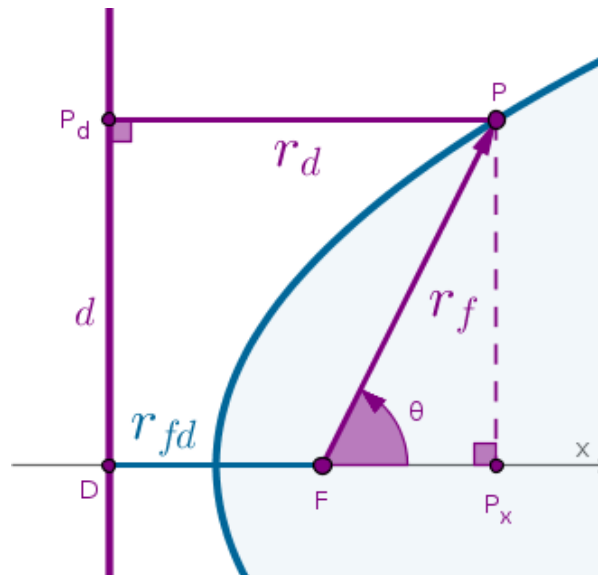
The Magic

We are now equipped with the minimum viable amount of mathematics to answer our main question, why do we place the origin of the polar frame at one of the curves' focal points?

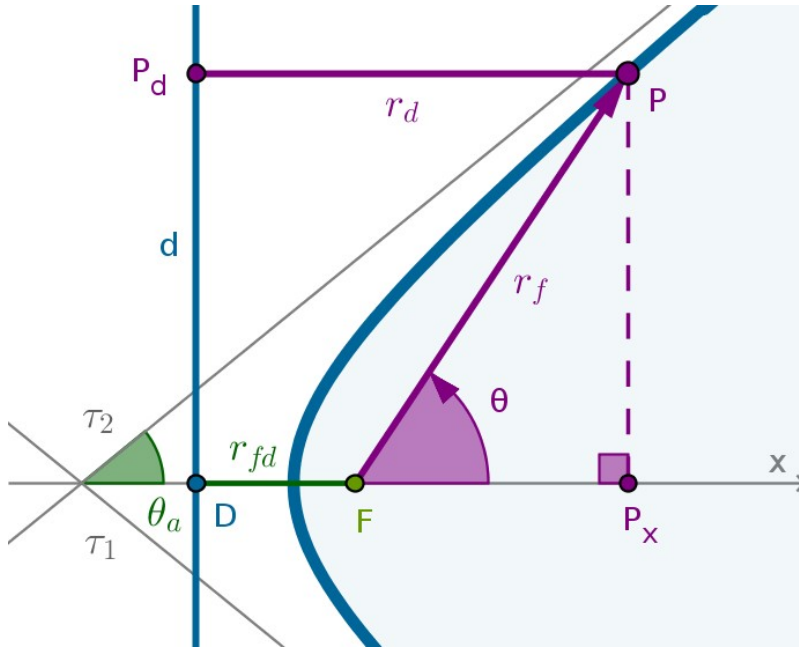
Even if we were not leading the respected audience through the garden path of the ready-made conclusions, purely from the art of problem solving perspective the number of expected failures is rather small because the usual suspects here are the curve's vertices and focal points, the Cartesian CS's origin, O , and, possibly, the points of intersection of the directrices with the focal axes.

It turns out that the focal points bring home the bacon. For an outline of a proof, place the polar frame's origin at:

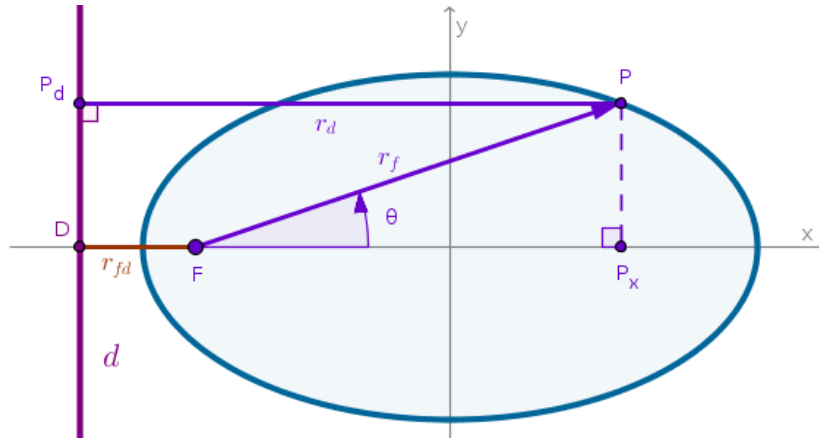
- the parabola's only focus, F (Figure 2.8):



- the hyperbola's right focus, F_2 , named F for consistency (Figure 2.9):



- the ellipse's left focus, F_1 , named F for consistency as well (Figure 2.10):



Then, thanks to all the legwork that we have done prior, for all three curves in unison we decompose the line segment DP_x as follows:

$$DP_x = DF + FP_x$$

$$r_d = r_{fd} + r_f \cdot \cos(\theta)$$

For the purpose of transforming the last relation, take r_d from $r_f/r_d = e$.

Take r_{fd} from $r_{fd} = p/e$.

Lose the subscript for r_f , which now plays the role of a radius vector, r :

$$\frac{r}{e} = \frac{p}{e} + r \cos(\theta) \quad \Longrightarrow \quad r = p + r \cdot e \cdot \cos(\theta)$$

from where:

$$r(1 - e \cos \theta) = p \quad \Longrightarrow \quad \boxed{r = \frac{p}{1 - e \cos(\theta)}}$$

where for a parabola $0 < \theta < 2\pi$, for an ellipse $0 \leq \theta < 2\pi$ (since $e \cdot \cos(\theta) \leq e < 1$) and for a hyperbola:

$$\cos(\theta) < \frac{1}{e} = \frac{a}{\sqrt{a^2 + b^2}} = \cos(\theta_a)$$

where θ_a is an acute angle between the hyperbola's asymptote and its focal axis, f . Consequently, for a hyperbola $\theta_a < \theta < 2\pi - \theta_a$.

Admittedly, the poor cousin Mr. Circle does not really fit the established framework. However, *a circle* can still be inscribed into the above family if we agree to define its focal parameter, p , as its radius, as in $p = r$, and define its eccentricity as zero: $e = 0$.

Chapter 3

Why Inverse Square

In this chapter, as we promised earlier, we will prove the first of two implications, IF ellipse THEN inverse square. More officially, we will now prove that if a planet traces an elliptic orbit with a star in one of the ellipse's two foci then the central force that is generated by the said star must be inversely proportional to the square of the distance between the centers of the star and the planet.

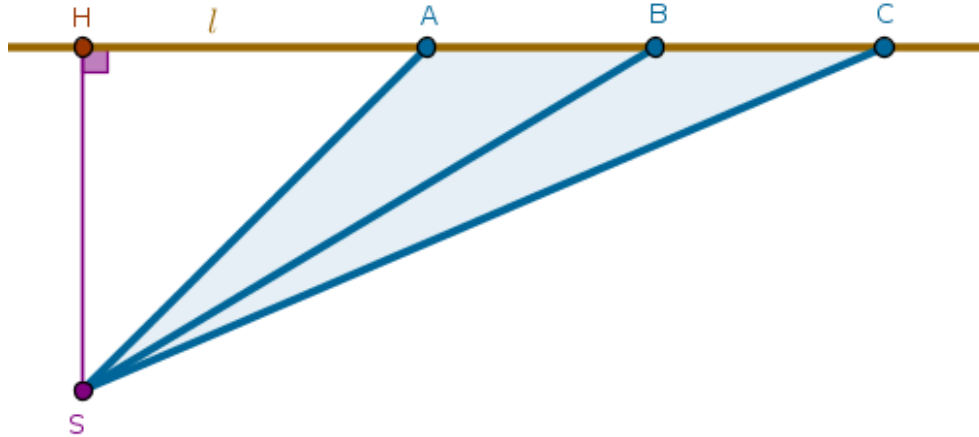
To that end, not only because we cannot possibly get into another person's mind, but mostly in order to avoid the idle speculation, below we present a plausible way of how *we* can arrive at the above theorem admitting, of course, that we are armed with the after-the-fact knowledge, a specific, highly technical and prolonged, education in the field and so on.

Moreover, below we enumerate the only laws and units of information that can be used in the upcoming deduction:

- [Euclid's "Elements"](#), because by the age of 23 Newton knew that work back and forth
- **G1**: Galileo's Law of Inertia, known to Newton and according to which an undisturbed body in motion will remain traveling at a constant speed along a straight line forever

- **K1, K2 and K3:** the three Kepler's Laws Of Planetary Motion, described earlier and also known to Newton
- Lunar orbital radius, $R_L \approx 3.8 \times 10^8 \text{ m}$, and its period of revolution around the Earth, $T_L = 28$ days
- Earth's radius, $R_E \approx 6400 \text{ km}$

Let a planet travel by the Sun, S , in a straight line, l , *in the absence of any forces* and let H be an orthogonal projection of S on l (Figure 3.1):



From **G1** it follows that in equal times the planet will cover equal distances:

$$|AB| = |BC| = v \cdot \Delta t \quad (1)$$

where v is the planet's constant velocity and Δt is the time it took the planet to travel from the point A to the point B and, likewise, from the point B to the point C .

Since the line segment SH is shared by $\triangle ASB$ and $\triangle BCS$ and the bases of these triangles are equal one another, from [Euclid's Book 1 Proposition 38](#) it follows that the areas of these two triangles are equal:

$$A_{\triangle ASB} = A_{\triangle BSC} \quad (2)$$

and **K2** certainly holds.

Next, like Newton, we assume that there is one, and only one, (central) force acting on the planet whose velocity, according to Galileo, will change. But where exactly will this force point to?

If the force acts along the straight line l then the direction of the planet's velocity will remain the same and the only parameter that will change is its magnitude: if the force points from the point A to the point B then the planet will accelerate and if the force points from the point B to the point A then the

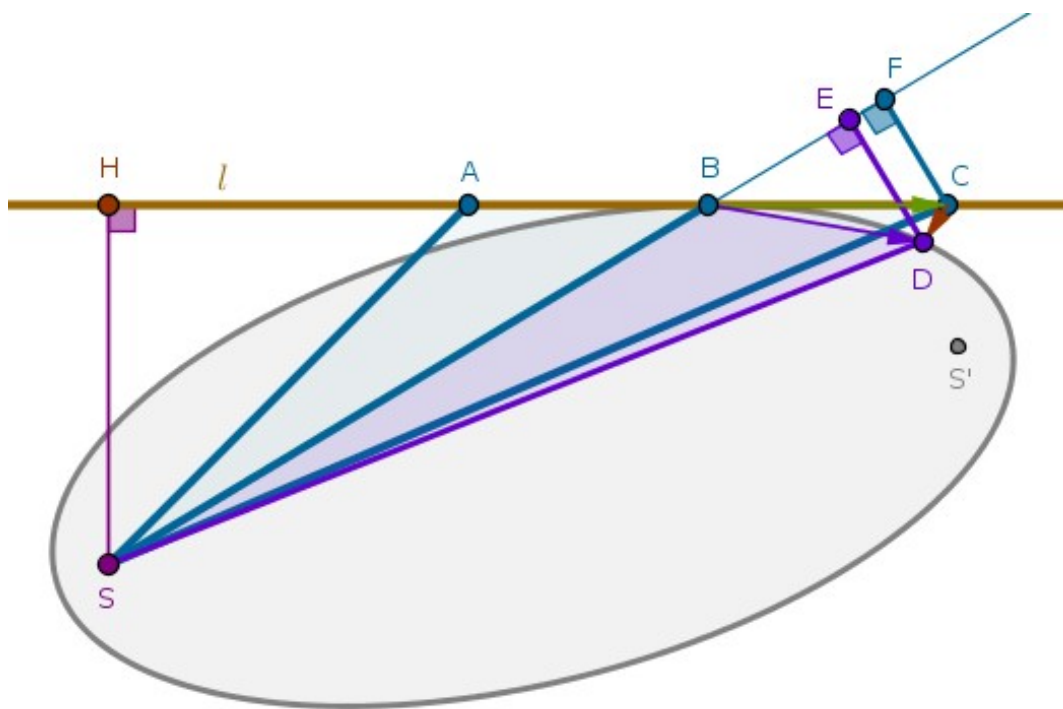
planet will slow down. In either case in equal times the planet will cover different distances, the triangles' bases and areas will no longer be equal and **K2** will be violated, which is a contradiction.

Hence, the force in question must point somewhere else. But where?

There is of course no way of telling exactly how but Newton came up with the following bright idea. In order to keep a planet in its elliptic orbit, there is no need for this tangential force along the straight line l to begin with. Why? Because according to **G1** the planet would have traveled along l on its own anyway, by inertia.

A thought experiment: if the Sun were to instantly disappear, meaning that the force of gravity would stop acting on a planet, in the time it takes the light to run from the Sun's former center to the planet's center the planet, in agreement with **G1**, would start running along the very straight line l , tangentially to its formerly elliptic orbit.

We know that according to **K1** the planet's motion does deviate from a straight line. In accordance with Henri Poincaré's saying that "geometry is an art of correct reasoning from incorrectly drawn figures", we now exaggerate a planet's elliptic orbit unrealistically for demonstration purposes and argue that instead of the point C where the planet would have been if it were to travel along a straight line, l , the planet is now at the point D on the circumference of the ellipse (Figure 3.2):



Consider the triangles BSD and $\triangle BSC$. A logical connection between the straight line l and the ellipse is that even though the shape of the trajectory changed, the areal velocity or the speed with which the area of the highlighted triangles changes remains the same, that much we require, by **K2**.

As such, from **K2** and (2) it follows that the areas of $\triangle BSD$ and $\triangle BSC$ must be equal. However, this time we look at the triangles a bit differently. Since they share the common side BS , we claim that BS is the common base which means that for the equality of the areas to hold the following equality must be true:

$$|DE| = |CF| \quad (3)$$

But from the same [Book 1 Proposition 38](#) it follows that these triangles must be “in the same parallels” which means that the straight line passing through the points C and D must be parallel to the straight line passing through the points B and S .

We now take off a hat that says “a geometer”, put on a hat that says “a physicist” and ponder, what exactly CD is, as a vector?

Early in “Principia” Newton shows the Parallelogram Law of Vector Addition so we shall argue from vectors also:

$$\overrightarrow{BC} + \overrightarrow{CD} = \overrightarrow{BD} \quad (4)$$

$$\overrightarrow{CD} = \overrightarrow{BD} - \overrightarrow{BC} \quad (5)$$

But, assuming that the point A also belongs to the circumference of an ellipse:

$$\overrightarrow{BC} = \overrightarrow{AB} \quad (6)$$

and:

$$\overrightarrow{AB} = \overrightarrow{v_{AB}} \cdot \Delta t \quad (7)$$

$$\overrightarrow{BD} = \overrightarrow{v_{BD}} \cdot \Delta t \quad (8)$$

where $\overrightarrow{v_{AB}}$ and $\overrightarrow{v_{BD}}$ are the *average* velocities of the planet between the corresponding points on its trajectory, we obtain:

$$\overrightarrow{CD} = (\overrightarrow{v_{AB}} - \overrightarrow{v_{BD}}) \cdot \Delta t \quad (9)$$

In other words $\overrightarrow{CD}$ is collinear with the vector that keeps track of the change of the average velocities of the planet. Since according to Galileo a force is required to change the velocity of a body, it is reasonable to introduce a new hypothesis that the (net) force acting on a body must be collinear with the direction of the change in this body’s velocity known as acceleration.

So far our (as well as Kepler’s) reasoning was loose because we have been comparing the area of the planar region bounded only by straight lines to the area of the planar region bounded by an arc of a

smooth curve which technically we are not allowed to do yet. In order to make the reasoning more rigorous, we see how the need for a new type of mathematics, calculus, arises. If we apply the $\Delta t \rightarrow 0$ limit to the difference of the planet's average velocities:

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t}$$

then the planet's *instantaneous* acceleration will be collinear with BS in a very special way for it will point *directly at* the Sun S , which is to say that such a force must be *central*.

Now that we have made a major breakthrough with respect to the presence and the direction of the central force, how can we deduce its magnitude? The answer is: from **K3**.

Assume that the force sought after, $F(r)$, is proportional to the acceleration:

$$F(r) = m \cdot a \quad (10)$$

This assumption is quite natural because according to Galileo, the force grows as acceleration grows and we already reached a conclusion about the collinearity of the vectors $\vec{F}$ and $\vec{a}$ (historically accurate is the formulation of (10) in terms of change of momentum or the “amount of motion” as Newton called it).

Next, we take another novel step. We claim that a very small portion of an arbitrary smooth curve can be approximated with a circular arc, in which case:

$$F(r) = m \cdot a = m \frac{v^2}{r} = m \frac{\omega^2 r^2}{r} = m \frac{4\pi^2 r}{T^2} \quad (11)$$

from where:

$$T^2 = m \frac{4\pi^2 r}{F(r)} \quad (12)$$

Divide both sides of (12) by r^3 :

$$\frac{T^2}{r^3} = m \frac{4\pi^2}{r^2 F(r)} \quad (13)$$

But according to **K3** we have:

$$\frac{T^2}{r^3} = k \quad (14)$$

and we argue that the above ratio, captured in k and known as Kepler's constant, is the same for all planets.

Which means that the magnitude on the right hand side of the equal sign in (13):

$$\frac{T^2}{r^3} = m \frac{4\pi^2}{r^2 F(r)} = k \quad (15)$$

must also be the same for all planets, which means that it cannot depend on r , which means that the product:

$$r^2 \cdot F(r)$$

cannot depend on r , which means that $F(r)$ must be inversely proportional to **the square** of the distance:

$$F(r) \propto \frac{1}{r^2}$$

or, from the last equality in (15):

$$\boxed{F(r) = 4\pi^2 \frac{m}{k \cdot r^2}} \quad (16)$$

How do we validate the relation shown in (16)?

One way to do so is to compare a known independently obtained numerical value of some physical magnitude to its theoretical equivalent, that of an acceleration of free fall, g , for example. If g_i is in a numerical agreement with g_t then may be, and only may be, we are on the right track, for on p. 376 of “*The Ultimate Quotable Einstein*” by Alice Calaprice we find that Mother Nature “... never says ‘yes’ to a theory, but only ‘maybe’ under the best of circumstances ...”

Hold on a second. How do we know that we should be comparing the accelerations of free fall, g ? Well, these accelerations are a side effect of another two stunning insights which hopefully show that the process of solving problems is not linear and smooth but rather choppy and quantum.

Our first insight is that even though **K3** is formulated for the star/planet pairs, it is nonetheless applicable to the planet/moon pairs in equal measure, with a potentially different ratio k . Since the needed magnitudes for the pair **Earth/Luna** are known, we obtain:

$$\frac{T_L^2}{R_L^3} = k_{EL} \approx 9.87 \cdot 10^{-14} \frac{\text{s}^2}{\text{m}^3} \quad (17)$$

Hm. Now what? The Earth does not have any more moons and we cannot compare the magnitude from (17) to anything else. But wait, what is this apple doing at our feet? How did it get here? Aha! Our second insight is that the force that makes the Moon fall on Earth and the force that makes an apple fall on Earth is one and the same!

Hence, our fearless prediction is that the theoretical acceleration of the free fall g_t from (10) must match its counterpart from (16) if we divide both sides of (16) by m :

$$\frac{F(r)}{m} = g_t = \frac{4\pi^2}{k_{EL} \cdot R_E^2} \quad (18)$$

where we used R_E for the distance between two bodies since the magnitude of R_E is way larger than the distance between the center of the apple and the Earth's surface:

$$g_t \approx 9.77 \frac{\text{m}}{\text{s}^2} \quad (19)$$

Numerically g_t from (19) agrees with the magnitude of $g_i \approx 9.8 \frac{\text{m}}{\text{s}^2}$ obtained independently (from Huygens' formula for the period of oscillation of an ideal mathematical pendulum, for example, or the Galileo ratios) and our deduction is complete.

It may come as shocking news to many but Newton did not use the form of the Universal Law of Gravity as it is taught in schools and colleges today. In "Principia" ratio, or fraction, was the first class citizen and using the modern symbol of $\propto$ or "proportional to", Newton's rendering of the law was:

$$F(r) \propto \frac{1}{r^2}, \quad F(r) \propto Mm \quad (20)$$

which was perfectly acceptable since in practical calculations that used astronomical units the magnitude of G , the gravitational constant, was not needed.

Even though later on, during the 18-th century, Johann Bernoulli, J. Lagrange and L. Euler translated Newton's bulky ratios into the modern compact analytic language of equations, we see the gravitational constant mentioned explicitly only in 1811 in S. D. Poisson's "Traité de mécanique" in Volume 1 under paragraph 242 on page 464:

$$\frac{fMm}{r^2}$$

where the symbol f played the role of G . Overall, the fate of G and its introduction into modern physics is, strangely enough, not very well understood. What can be stated with a reasonable degree of certainty is that being the author of "for every action there is an equal and opposite reaction", Newton eventually realized that if the Sun acts on a planet then the planet must also act on the Sun which is, possibly, how the product of two masses made it into (20). And, using the modern knowledge, from:

$$G \frac{Mm}{r^2} = 4\pi^2 \frac{m}{k \cdot r^2}$$

we obtain:

$$k \cdot G \cdot M = 4\pi^2$$

where k is the Kepler's constant for the pairs Star/Planet, G is the gravitational constant and M is the mass of the Star.

More can be said on the subject but even at this point it is safe to claim that Tyge Brache did not live in vain if only because, in a circuitous path from experiment to theory and back to reality again, all three Kepler's laws can be deduced from Newton's Law of Gravity.

In addition, though technically planets' orbits are a capricious and fussy, if not messy, business, there exists another amazing feat of abstract thinking that is nothing short of breathtaking.

If we start out with simple geometric definitions of the three conical sections, which seemingly have nothing to do with physics, then they all have the same analytical representation or an algebraic equation which happens to be the solution of a differential equation that describes the motion of the celestial bodies, the equation in which Newton's Law of Gravity plays the central, no pun intended, role.

And that is the matter of the next chapter.

Chapter 4

Why Ellipse

In this chapter we will prove the second of two implications, IF inverse square THEN ellipse. More officially, we will now prove that if a planet is acted on by a single (central) gravitational force then that planet will trace an elliptic orbit with a star in one of the ellipse's two foci. It should be understood that in this case the star plays the role of the center of force.

To that end, we, considerably, simplify our task by making the following explicit assumptions:

- the solar system has one and only planet
- the Sun's mass, M , is infinitely larger than the planet's mass, m
- the Sun is stationary
- the masses m and M remain constant at all times
- both celestial bodies are represented by material points or point masses
- the velocities in question are way, way smaller than the speed of light in vacuum
- space curvature due to M is ignored

All of these assumptions do have a technical merit behind them. First off, in reality our solar system, of course, has more than one planet in it.

Next, the mass of the Sun is way, way larger than the mass of, say, Earth but mathematically the ratio of these two numbers is not zero.

Next, the Sun is actually not stationary because its center orbits the common center of mass of all the planets in the solar system.

The mass of the Sun and the mass of the Earth are not really constant, as these masses do change over time and so on.

In any case, in a lightning round, inevitably omitting some details which the reader is encouraged to recover on his/her own, the central force of gravity is conservative and, on the one hand, the work, W , that it performs in order to displace the mass m from infinity into r units of distance toward the star M is:

$$W = \int_r^\infty \frac{GMm}{r^2} dr = \frac{GMm}{r}$$

and on the other hand, that work is performed at the expense of the loss of potential energy U :

$$W = U_\infty - U_r$$

Taking U_∞ to be zero, we obtain for $U(r)$:

$$U(r) = -\frac{GMm}{r}$$

Since we neglect the kinetic energy of the Sun in comparison with the kinetic energy of the planet, from the law of conservation of energy for our system we have:

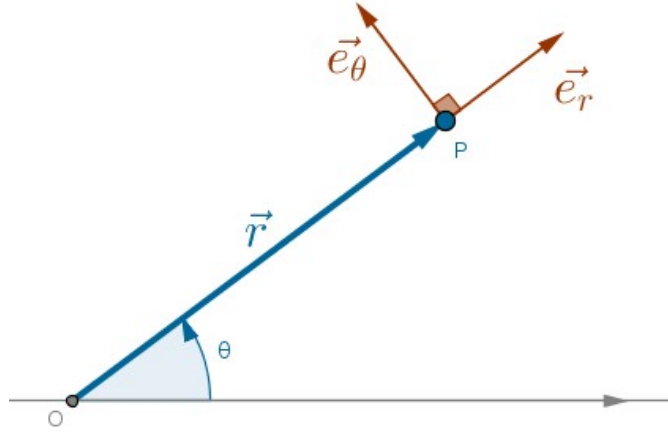
$$\frac{mv^2}{2} - \frac{GMm}{r} = E = \text{const} \quad (21)$$

Divide (21) through by m :

$$\frac{v^2}{2} - \frac{GM}{r} = \frac{E}{m} = \epsilon = \text{const} \quad (22)$$

Here ϵ is the planet's unit mass energy, $\vec{r}$ is the planet's radius vector with the Sun as its origin and $\vec{v}$ is the planet's velocity which we decompose into two mutually orthogonal vectors, $v_r \cdot \vec{e}_r$, the radial velocity, which will keep track of the rate of change of the radius vector r , and $v_\theta \cdot \vec{e}_\theta$, the tangential velocity, which will keep track of the rate of change of the planet's angular displacement at r units of distance from M with respect to O and the polar axis, in polar coordinates (r, θ) .

We will measure the angular displacement θ in the traditional positive or counterclockwise direction (Figure 4.1):



where:

$$\vec{v} = v_r \vec{e}_r + v_\theta \vec{e}_\theta \quad (23)$$

$$v_r = \dot{r} \quad (24)$$

$$v_\theta = r\dot{\theta} \quad (25)$$

From the Pythagorean theorem (or the Euclidean norm) for the square v^2 we have:

$$v^2 = v_r^2 + v_\theta^2 \quad (26)$$

Using (24) and (25), put v^2 from (26) into (22):

$$\boxed{\frac{\dot{r}^2 + r^2 \dot{\theta}^2}{2} - \frac{GM}{r} = \epsilon} \quad (27)$$

Since our force is central, on the one hand, the net torque in the system, defined as:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

is zero because the angle between $\vec{r}$ and $\vec{F}$ is π radians and $\sin \pi = 0$.

Thus:

$$\vec{\tau} = \vec{r} \times \vec{F} = 0 \quad (28)$$

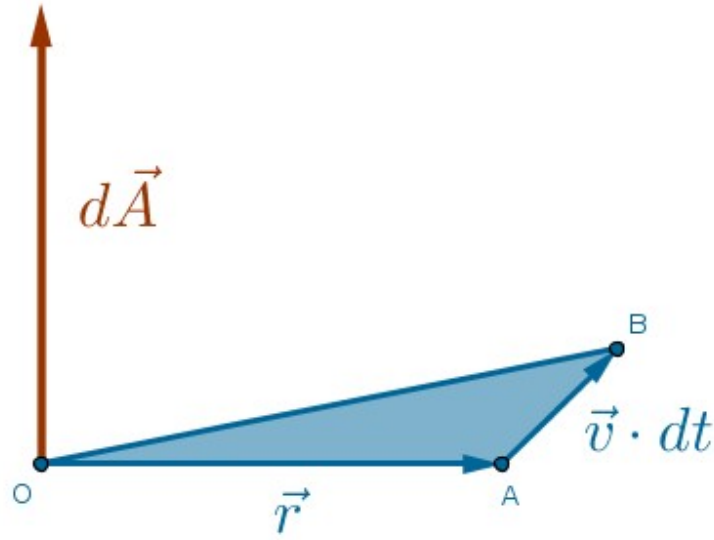
On the other hand, the torque as the first derivative of the angular momentum, $\vec{L}$, over time:

$$\vec{\tau} = \frac{d\vec{L}}{dt} \quad (29)$$

must also be zero since the left hand sides of the equal signs in (28) and (29) are identical:

$$\vec{\tau} = 0 = \frac{d\vec{L}}{dt}$$

which means that since the derivative of a function is zero, the function itself, $\vec{L}$, must remain constant over time and in our case $\vec{L}$ has a simple geometric interpretation. Assuming that over time dt the point m was displaced by $\vec{v} \cdot dt$ (Figure 4.2):



the area $d\vec{A}$ of the colored triangle in the vector form is:

$$d\vec{A} = \frac{1}{2} \vec{r} \times \vec{v} dt$$

or:

$$\frac{d\vec{A}}{dt} = \frac{1}{2} \vec{r} \times \vec{v} \quad (30)$$

where $\dot{\vec{A}}$ is the particle's areal velocity.

Since we agreed that the mass m is constant, for the planet's angular momentum we have:

$$\vec{L} = \vec{r} \times \vec{p} = m \vec{r} \times \vec{v} \quad (31)$$

and we see that the magnitudes on the left hand sides of the equal signs in (30) and (31) differ only by a coefficient and, hence, we can easily manufacture one from the other:

$$\vec{L} = 2m \vec{A}$$

but since m is constant and $v \ll c$, from the constancy of $\vec{L}$ the constancy of $\vec{A}$ follows, which also implies that the motion is strictly planar because the vectors $\vec{r}$ and $\vec{v}$ remain orthogonal to $\vec{A}$ at all times.

Name, for brevity, the areal velocity s , for **sector**, observe that it is a constant quantity for our planet and use (23), (24) and (25) to compute it:

$$s = \frac{1}{2} \vec{r} \times \vec{v} = \frac{1}{2} (\vec{r} \times \vec{e}_r v_r + \vec{r} \times \vec{e}_\theta v_\theta)$$

The cross or vector product of $\vec{r} \times \vec{e}_r$ is zero because these vectors are collinear and hence:

$$s = \frac{1}{2} \vec{r} \times \vec{e}_\theta v_\theta = \frac{1}{2} r^2 \dot{\theta} = \text{const}$$

or:

$$\boxed{\frac{1}{2} r^2 \dot{\theta} = s} \quad (32)$$

and we need to solve two differential equations, (27) and (32), for two variables, r and θ .

The way it sits right now the equation (27) is not linear but with a proper substitution it can be reduced to something rather trivial. Namely. From (32) we have for $\dot{\theta}$:

$$\dot{\theta} = \frac{2s}{r^2} \quad (33)$$

Since we seek r as a function of θ , with (33), we have:

$$r = r(\theta), \quad \dot{r} = \frac{dr}{d\theta} \dot{\theta} = \frac{dr}{d\theta} \frac{2s}{r^2} \quad (34)$$

Put (33) and (34) into (27):

$$\left(\frac{1}{r^2} \frac{dr}{d\theta}\right)^2 + \frac{1}{r^2} = \frac{1}{2s^2} \left(\epsilon + \frac{GM}{r}\right) \quad (35)$$

Introduce a new variable, ρ , with the following substitution:

$$\rho = -\frac{1}{r} + \frac{1}{p}, \quad p = \text{const} \quad (36)$$

where p is our friend focal parameter and the idea of using it in (36) is to eliminate the first powers of ρ after it is put into (35):

$$\left(\frac{d\rho}{d\theta}\right)^2 + \left(\rho - \frac{1}{p}\right)^2 = \frac{\epsilon}{2s^2} + \frac{GM}{2s^2} \left(-\rho + \frac{1}{k}\right) \quad (37)$$

Group all the first powers of ρ in (37) and require that its coefficient must be zero:

$$\rho \left(\frac{GM}{2s^2} - \frac{2}{p}\right) = 0 \quad (38)$$

from where:

$$p = \frac{4s^2}{GM} \quad (39)$$

With the magnitude of p from (39) the equation (37) collapses into:

$$\left(\frac{d\rho}{d\theta}\right)^2 = \frac{\epsilon}{2s^2} + \frac{1}{p^2} - \rho^2 \quad (40)$$

Since the first two terms on the right hand side of the equal sign in (40) are not negative and constant, we may, for brevity, designate them as follows:

$$\frac{\epsilon}{2s^2} + \frac{1}{p^2} = C^2, \quad C = \text{const} \quad (41)$$

Then (40) becomes:

$$\left(\frac{d\rho}{d\theta}\right)^2 = C^2 - \rho^2 \quad (42)$$

But the left hand side of (42) is a square of a real number and hence:

$$C^2 - \rho^2 \geq 0 \quad (43)$$

and we have the right to introduce our last substitution:

$$\rho = C \cos(\omega) \quad (44)$$

after which (42) becomes:

$$C^2 \sin^2(\omega) \frac{d\omega}{d\theta} = C^2 \sin^2(\omega) \quad (45)$$

or (as promised):

$$\frac{d\omega}{d\theta} = \pm 1 \quad (46)$$

from where $\omega = \pm\theta + \theta_0$ and $\rho = C \cos(\theta \pm \theta_0)$ since the cosine is an even function. If we designate:

$$e = C \cdot p = \sqrt{1 + \frac{\epsilon p^2}{2s^2}} = \sqrt{1 + \frac{8\epsilon s^2}{G^2 M^2}} \quad (47)$$

unwind our substitutions and put $\theta_0 = 0$ then:

$$\boxed{r(\theta) = \frac{p}{1 - e \cos(\theta)}} \quad (48)$$

where our choice for the angle θ_0 simply means that the initial state of motion of a planet corresponds to such a position of that planet that the length of its radius vector is:

$$r_0 = \frac{p}{1 - e}$$

Either way, in the previous chapter we have seen that the equation (48) encodes exactly a conic section, which, in general is what was required to prove. $\square$

We observe in passing that the actual planetary orbits is a much more involved and fussy business because a cosmic body that either does not have or has and does not engage its own means of propulsion and moves solely under the influence of the central force of gravity can transcribe the following trajectories:

- a straight line that passes directly through the parent star
- a hyperbola
- a parabola
- an ellipse or
- a circle

and all these claims can be proved but such proofs are beyond the scope of this small book and we hope the amount of technical information that we present in this book both paints an adequate picture that

explains why planetary orbits are what they are and is enough to stimulate our reader's desire to investigate the matters of classical celestial mechanics further.

Chapter 5

An Isomorphism In Celestial Mechanics

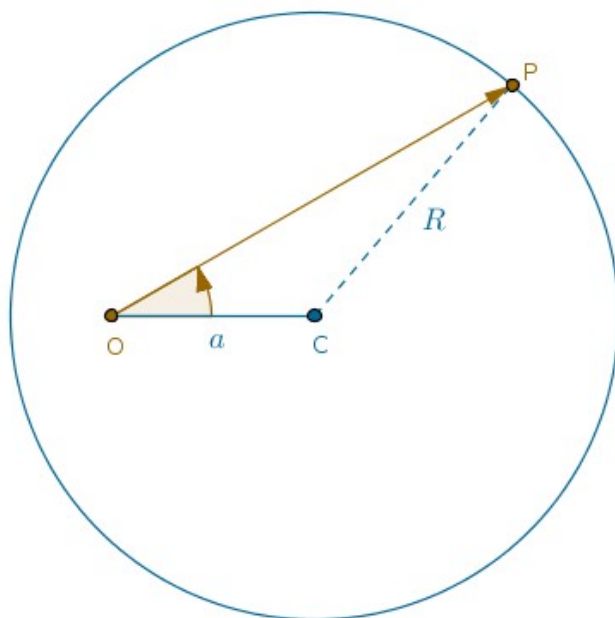
In this, final, chapter of the book we will solve a beautiful problem that starts out as an innocent and, perhaps, even boring “yet another exercise in kinematics” but which, under a proper interpretation, quickly becomes and blossoms into an amazingly harmonious tandem of mathematics and physics. In order to discover that interpretation, we need to be aware of exactly the material that we have covered in the first four chapters.

The goal of this chapter is to give our readers an even more realistic and tangible opportunity to deal with the very basic and introductory material from classical celestial mechanics that is already unburdened from matters historical and that is cast in a modern technical language.

If the mathematical apparatus used in this chapter seems to be beyond the current level of any one of our readers then we suggest that these readers skim through this chapter anyway and nonetheless, make a mental note of it, keep the spark of interest burning and come back to this material later on, when the technical devices that we use in this chapter become more accessible and familiar.

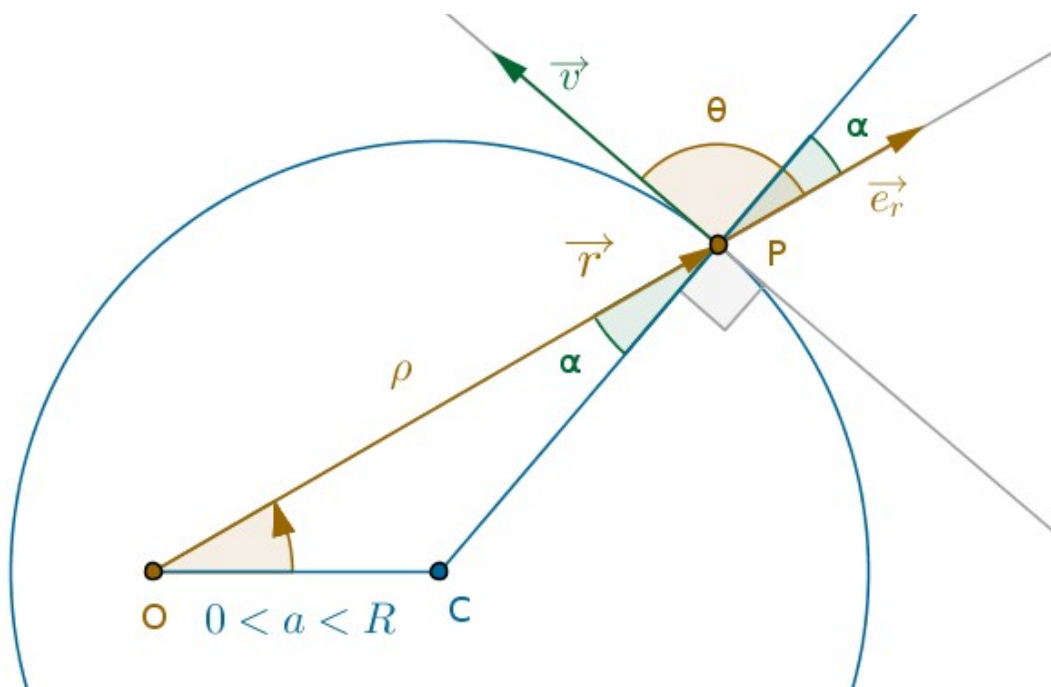
We also suggest that our readers try to solve the problem below by themselves first, before looking at our presentation.

Problem 1. A point, P , is moving along the periphery of a circle of radius R with a constant areal velocity σ_0 with respect to a point O that is located in the same plane $0 < a < R$ units of distance away from the center of the circle, C (Figure 5.1):



Find: the velocity, v , of the moving point as a function of the distance $\rho = |OP|$.

Solution. Observe that the vector of the moving point's velocity, $\vec{v}$ (Figure 5.2):



remains tangent to the point's trajectory, which is a circle. Hence, that vector remains perpendicular to the radial direction CP at all times. The areal velocity of a point P , in general, is given by:

$$\vec{\sigma} = \frac{1}{2} [\vec{r} \cdot \vec{v}]$$

Represent the radius vector $\vec{r}$ of the point P with respect to the point O via the radial unit vector $\vec{e}_r$:

$$\vec{r} = \rho \vec{e}_r, \quad \rho = |OP|$$

Then, the above areal velocity becomes:

$$\vec{\sigma} = \frac{1}{2} [\vec{e}_r \cdot \vec{v}] \rho$$

Therefore, the projection of that areal velocity onto the z -axis will be:

$$\sigma_z = \frac{1}{2} \rho v \sin(\theta) = \sigma_0$$

where θ is the angle between the velocity v of the point P and the respective radial unit vector e_r . From the above, we have for the said velocity $v(\rho)$:

$$v(\rho) = \frac{2\sigma_0}{\rho \sin(\theta)}$$

From the triangle OCP , by the Law of Cosines, we have:

$$a^2 = \rho^2 + R^2 - 2\rho R \cos(\alpha)$$

But for the angle between the velocity v of the point P and the respective radial unit vector e_r we have:

$$\theta = \frac{\pi}{2} + \alpha \implies \cos(\alpha) = \cos\left(\theta - \frac{\pi}{2}\right) = \sin(\theta)$$

Hence:

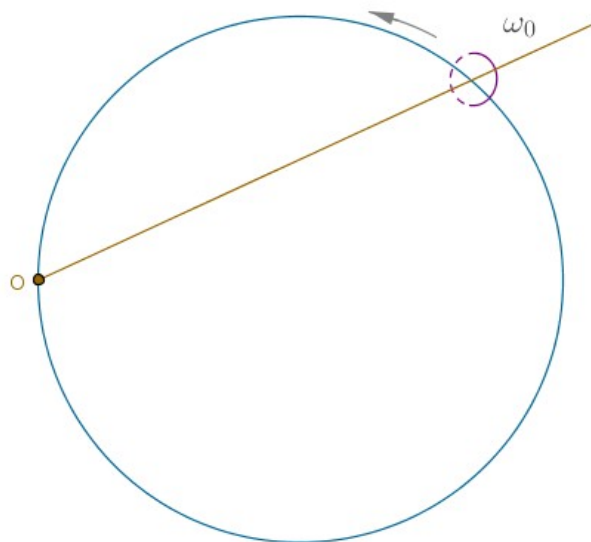
$$\cos(\alpha) = \sin(\theta) = \frac{\rho^2 + R^2 - a^2}{2\rho R}$$

and the game is over:

$$v(\rho) = \frac{4\sigma_0 R}{\rho^2 + R^2 - a^2}$$

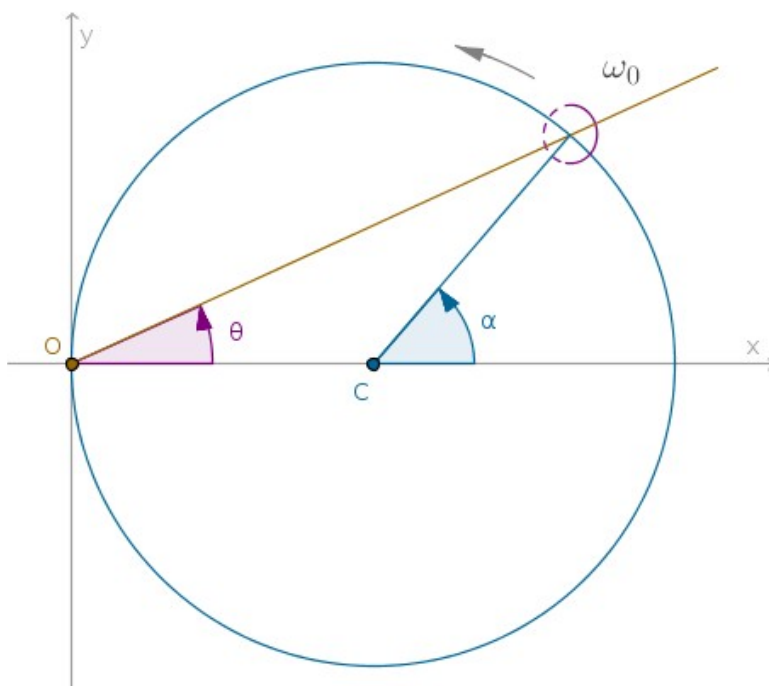
Problem solved. $\square$

Improvising on the given theme, we may, as well, invent the following related, more or less practical, problem. Imagine that on a thin wire in the shape of a circle of radius R there sits a small ring through which there passes a rod that rotates about a point O on the circle in the circle's plane with a constant angular velocity ω_0 (Figure 5.3):



Find: the velocity and the acceleration of the ring.

Solution. In this problem instead of the areal velocity, we are given the angular velocity of the rod with respect to the point O (Figure 5.4):



If we place the origin of a Cartesian coordinate system precisely into that point then, as the rod rotates in the parent plane, counterclockwise, it will form a certain angle, θ , with the positive branch of the chose x -axis and for the angular velocity of that rod we will have:

$$\omega_0 = \frac{d\theta}{dt} \implies \int_{\theta_0}^{\theta} d\theta = \int_0^t \omega_0 dt = \omega_0 t$$

If we agree that at the initial moment of time $t = 0$ the above initial angle θ_0 was 0 then we shall have:

$$\theta(t) = \omega_0 t$$

For the linear velocity v of the small ring in terms of its angular velocity $\dot{\alpha}$, we have:

$$v = \omega R = \dot{\alpha} R$$

But by [Euclid's "Elements" Book 3 Proposition 20](#), in a circle the angle at the center is double the angle at the circumference. Hence, $\alpha = 2\theta$ and:

$$v(t) = \dot{\alpha} R = 2\dot{\theta} R = 2\omega_0 R = \text{const}$$

Therefore, since the magnitude of the ring's velocity is constant, it follows that the tangential acceleration of that ring will be zero.

Since a complete acceleration of a moving object is a vector sum of the tangential and the normal accelerations and we established that the tangential acceleration of the ring is zero, it follows that the only component that will make a non-zero contribution to the ring's acceleration will be the normal acceleration of the ring:

$$a = a_n = \frac{v^2}{R} = 4\omega_0^2 R = \text{const}$$

If we step further and ask ourselves, what is the ring's areal velocity relative to the rod's pivot point, O , and what is the ring's angular velocity, then ...

We instantly see that according to the definition of angular velocity, for our ring we have:

$$\omega(t) = \frac{d\alpha}{dt} = \frac{d(2\theta)}{dt} = \frac{d(2\omega_0 t)}{dt} = 2\omega_0 = \text{const}$$

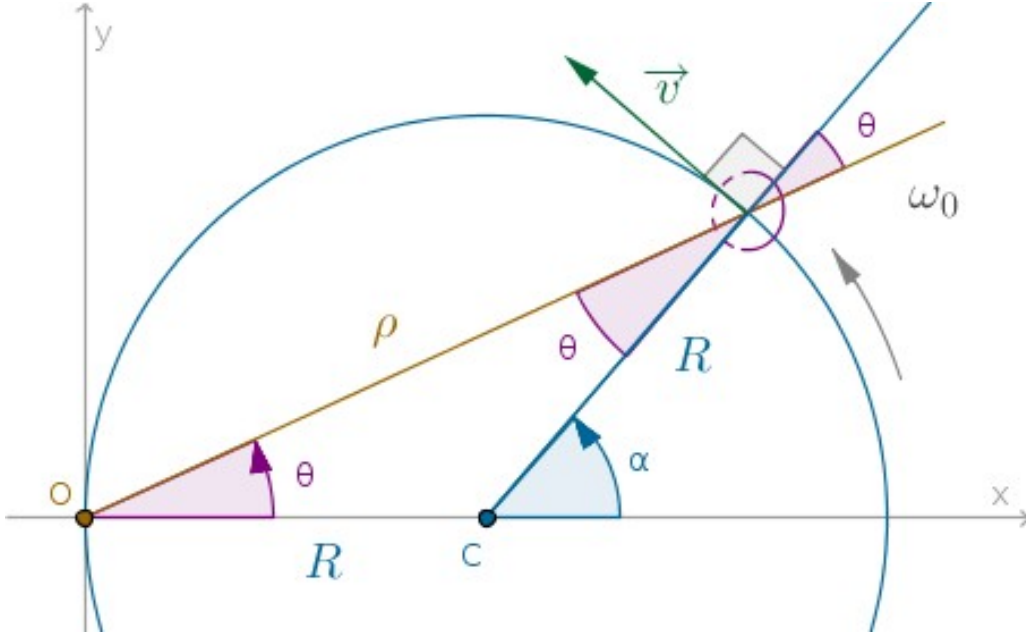
In other words, the angular velocity of the small ring, in this problem, is a quantity that is constant.

And what about the ring's areal velocity relative to the points C and O ?

Well, the ring's areal velocity, σ_z^c relative to the point C is more than straightforward:

$$\sigma_z^c = \frac{1}{2} \cdot R \cdot v \cdot \sin\left(\frac{\pi}{2}\right) = \omega_0 R^2 = \text{const}$$

Aha. The areal velocity of the ring relative to the circle's center is, yet again, a quantity that is constant. In order to compute the areal velocity of the ring relative to the pivot point O (Figure 5.5):



we, essentially, do the same as in the original problem:

$$\sigma_z^o = \frac{\rho}{2} \cdot v \cdot \sin\left(\frac{\pi}{2} + \theta\right) = \rho \cdot \omega_0 R \cos(\omega_0 t)$$

because $v = 2\omega_0 R$. Since the triangle OC Ring is isosceles, it follows that $2R \cos(\omega_0 t) = \rho$ and, thus:

$$\sigma_z^o(t) = 2\omega_0 R^2 \cos^2(\omega_0 t) \neq \text{const}$$

and we see that this time the areal velocity of the ring relative to an off-center point is *not* constant. We know go back to our original problem and think. Hm.

In that original problem the areal velocity of the orbiting point about an off-center point **was** constant, while in its improvisation sibling it is **not**.

Likewise, if the angular velocity of the orbiting object, a ring, in the sibling problem **was** constant then what will the corresponding angular velocity of the orbiting point be in the original problem?

	Original problem	Sibling problem
Areal Velocity	constant	variable
Angular Velocity	?	constant

$|CP| = R$
 $|CA| = a \cdot \cos(\theta)$
 $|AP| = \rho \cdot \cos(\alpha)$

$$\sigma_z^o = \frac{1}{2} \cdot \rho \cdot v \cdot \sin\left(\frac{\pi}{2} - \alpha\right) = \sigma_0$$

49

For the trigonometric function of sine we, trivially, have:

$$\sin\left(\frac{\pi}{2} - \alpha\right) = \cos(\alpha)$$

Thus, so far, for the areal velocity σ_0 , we conveniently have:

$$\sigma_0 = \frac{R}{2} \cdot \omega_c \cdot \rho \cdot \cos(\alpha)$$

where the rightmost projection of the distance ρ onto the radius of the parent circle, R , is recovered trivially from the fact that that radial distance, $|CP|$, is a quantity that is composite:

$$R = |CP| = |CA| + |AP| = a \cdot \cos(\theta) + \rho \cdot \cos(\alpha)$$

Thus:

$$\rho \cdot \cos(\alpha) = R - a \cdot \cos(\theta)$$

Hence:

$$\sigma_0 = \frac{R}{2} \cdot \omega_c \cdot (R - a \cdot \cos(\theta))$$

Solving the above equation for the angular velocity ω_c , which, by the way, by definition, is the first time derivative of the angle θ , we find:

$$\omega_c = \frac{2\sigma_0}{R \cdot (R - a \cdot \cos(\theta))} = \frac{d\theta}{dt}$$

Separate the variables in the above ODE:

$$2\sigma_0 dt = R \cdot (R - a \cdot \cos(\theta)) d\theta$$

Integrate its both sides, from 0 to t on the left and from 0 to θ on the right:

$$2\sigma_0 \int_0^t dt = R \cdot \int_0^\theta (R - a \cdot \cos(\theta)) d\theta$$

and obtain the following result:

$$2\sigma_0 t = R^2 \theta - aR \sin(\theta)$$

Divide that result through by a single copy of R^2 :

$$\theta - \frac{a}{R} \sin(\theta) = \frac{2\sigma_0}{R^2} t \quad (49)$$

and arrive at an equation that is transcendental over θ and t , meaning that there does not exist a finite combination of elementary functions that expresses the angle θ as an explicit function of the time t .

Consequently, the answer to our question of what the angular velocity of the orbiting point P is, is the following pair of equations:

$$\omega_c(t) = \frac{2\sigma_0}{R \cdot (R - a \cdot \cos(\theta(t)))} \quad (50)$$

and:

$$\theta(t) - \frac{a}{R} \sin(\theta(t)) = \frac{2\sigma_0}{R^2} t \quad (51)$$

and we see that the angular velocity $\omega_c(t)$ of our point P in the original problem is **not** constant! But our findings go deeper than that. If in the equations (48) and (51) we put:

$$e = \frac{a}{R}, \quad \theta = E, \quad M = \frac{2\sigma_0}{R^2} t \quad (52)$$

then these equations become [the classical Kepler's Equation](#):

$$E - e \cdot \sin(E) = M$$

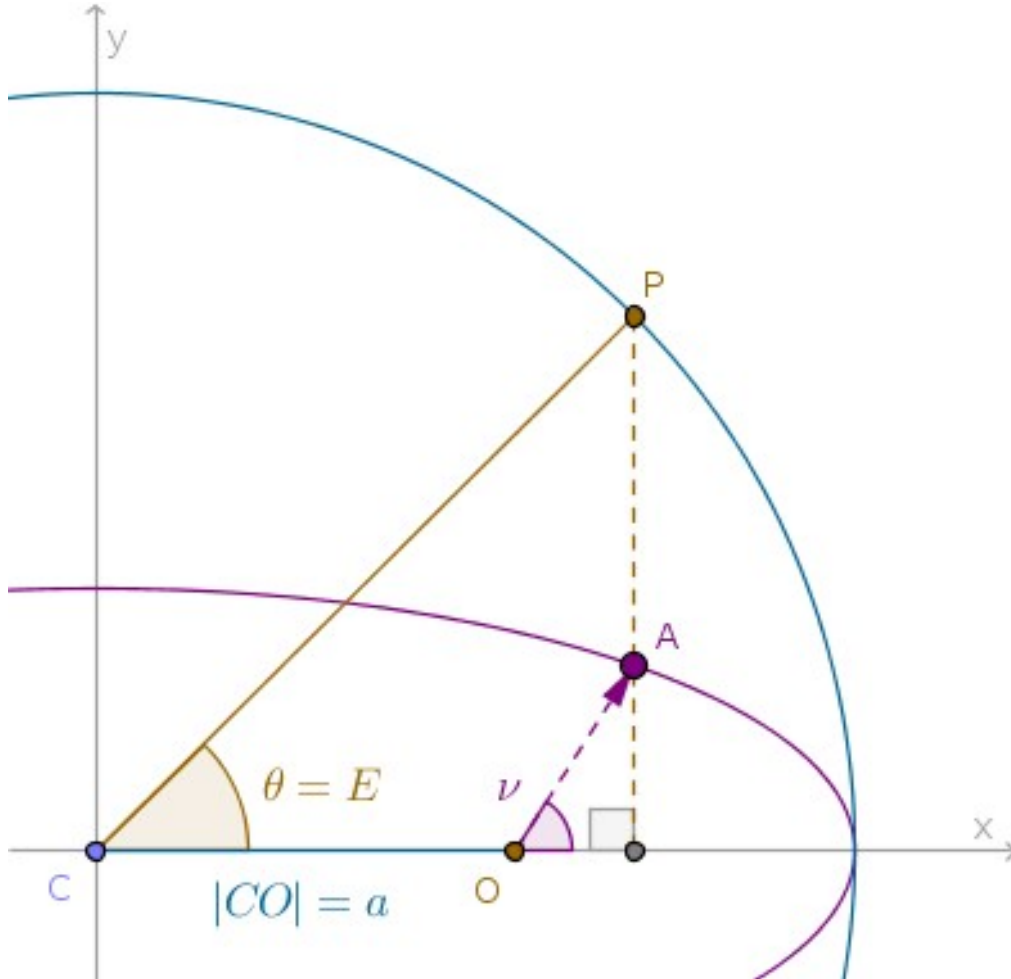
where M is the “mean anomaly”, E is the “eccentric anomaly” and e is the “eccentricity” and our extended original problem dovetails into the Kepler's Second Law in particular and the classical celestial mechanics in general as follows.

In an appropriately chosen Cartesian coordinate system construct an ellipse whose center, C , is located precisely at the origin.

Such an ellipse will represent the orbit of a planet about a star that is located in one of the foci of that ellipse.

Next, construct a circle whose center is also C and whose radius is equal to the larger semi-axis of the above ellipse.

Next, project our sample point P from the above circle onto the x-axis of the chosen coordinate system. The respective straight line of that projection will intersect the parent ellipse, in general, at two distinct points but we are only interested in one such intersection point, A (Figure 5.8):



It turns out*, see below, that the said point A represents the **actual** planet that orbits (the common center of mass with) the star at O .

In other words, our point P is the location where the actual planet A would have been were it to orbit its star in a perfectly circular orbit.

Our angle $OCP = \theta = E$ is exactly the so-called *eccentric anomaly* of the orbit. Our version of M , the mean anomaly of the orbit, is a uniformly growing angle **as if** the planet's orbit were perfectly circular and **as if** such a planet moved over such a circular orbit with a constant angular velocity $2\sigma_0/R^2$ (which it does not).

*Our circle is called **the reference circle** and such a circle can be bijected onto the underlying ellipse thus establishing a bijection between that circle and that ellipse, see above.

Namely, either the reference circle can be continuously deformed into the underlying ellipse by a uniform and continuous *compression* of the parent plane along the y -axis only or, equivalently, the said ellipse can be continuously deformed into the bounding reference circle by a uniform and continuous *expansion* of the parent plane along the y -axis only.

No transformation along the x -axis is needed in this case.

As such, to every point, P , on a circle there corresponds a particular point, A , on the ellipse and conversely, to every point on the ellipse, A , there corresponds a particular point, P , on the circle and our construction of the point A shown in Figure 8 above shows the above expand/compress continuous transformation intuitively.

Lastly, because the equation of motion (3) of our point P on a circle with a constant areal velocity relative to an off-center point is structurally the same as the equation of motion of a planet over an elliptic orbit, it follows that these motions are *isomorphic* or *structurally indistinguishable*.

The point O , where the star is, is called **the orbital focus**. The angle ν in Figure 5.8 is the so-called **true anomaly** or the true and actual angular displacement of the orbiting planet relative to the Sun which tells us where the planet actually is and so on and one of the lessons that we learn in this discussion is the following: a seemingly trivial problem in planar geometry encodes several hundred years of mathematical battles that lead to the development of classical celestial mechanics. $\square$

In summary, we started solving Problem 1 as if it were yet another, mundane, kinematics problem. Then, we showed a healthy amount of initiative by creating a problem that we thought was similar to Problem 1. In the long run it turned out not to be the case and in its spirit our improvisation on the theme was a turn in a wrong direction entirely.

However, eventually we managed to ask the right question, what is the angular velocity of the point in Problem 1?

While answering that question we stumbled upon a technical fact that the motion of the point in Problem 1 over a circle is structurally identical or isomorphic to the motion of a planet over an elliptic orbit with the Sun in one of its two foci.

Isomorphism in mathematics is a very powerful and versatile instrument. For example. Assume, for demonstration purposes, that while working on a certain problem we came across a group that is descriptively very different from a group that we know and love but later on we realize that that seemingly new group is structurally the same or isomorphic to a group that we know and understand well.

Then, any results that can be deduced for a well-known and well-understood group based only on group axioms can be lifted verbatim, shifted and dropped onto a new isomorphic group, where these results will simply be properly adjusted for the local material.

Likewise, isomorphism in physics is an equally powerful tool and for the same reason. If it can be shown that two, not necessarily mechanical, physical systems are isomorphic with respect to each other then all the relevant results that are legally obtained for one such system can be lifted verbatim, shifted and dropped onto an isomorphic buddy system with an appropriate adjustment of these results for the local material.

For example, instead of studying the motion of a certain point mass over an ellipse, it may prove to be more computationally feasible to study the motion of such a particle over an isomorphic orbit, such as a circle. And after certain results are obtained for a circular orbit, such results may be transferred onto an isomorphic elliptic orbit, where obtaining such results directly may be way, way more difficult.

